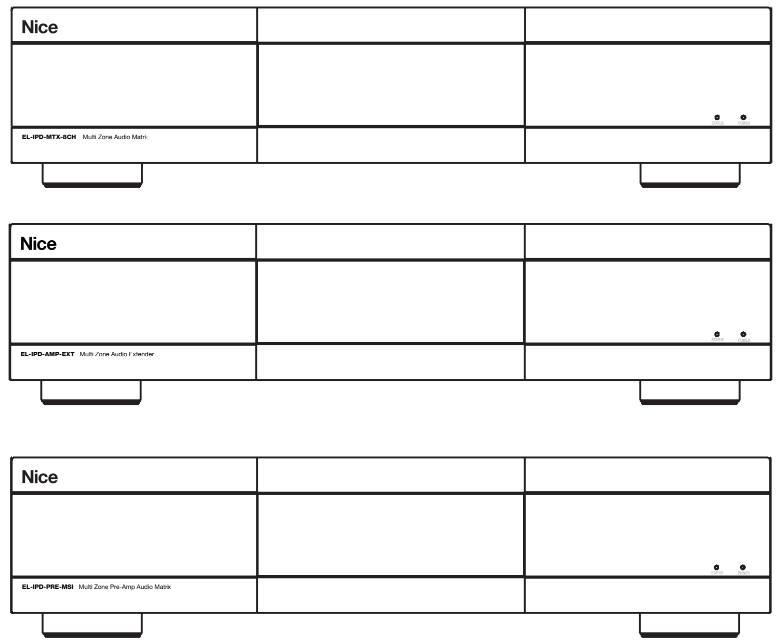
IP AMP DESIGN GUIDE



Flexible Audio Distribution

Design Guide

Nice

Table of Contents

System Overview 3
Designed to Meet Your Needs in Any Room 4
Multi Zone Audio Matrix - Deep Dive 5
Applications 8
Functionality Features 19
Zone Output Configuration 19
Audio Ramp 19
Mute Rules 19
Volume Curve 19
Equalizer 19
Expandability 19
Zone Specific Volume Parameters 19
Configuration Basics 20
Single vs multi chassis 20
Adding Chassis' to the Configurator 20
Configuring IPD devices 20
Resources 22
Sources 23
Zones 24
Configuring Zones 24
Maximizing Zones 27
Speaker Configuration 27
Pre Out Configuration 29
Multiple Chassis Configuration - Dante default (unicast) 30
Unicast overview 30
Advanced Multiple Chassis Configuration – Dante Multicast 34
Multicast Overview 34
Common Issues 37
Best Practices 38
Network setup and deployment: (unicast and multicast) 38
Networking Details 41
Sample Dante Multicast Configuration 42

System Overview

Whether your audio distribution project is large or small, the Nice Home Management IP amplifier products provide unprecedented flexibility. Nice Home Management IP enabled audio distribution simplifies audio distribution and enhances the intelligent property by delivering high quality audio over standard networks. Easily scaling from a single chassis solution to a decentralized multi-chassis / multi-source implementation, Nice Home Management IP audio distribution meets the needs for projects of just about any size. Because audio is delivered over the network, Nice Home Management IP amplifier products let you place sources and amplifiers where they make the most sense.

This is a complex and detailed document, covering many aspects from design to configuration of the Nice Home Management IP amplifier products. How they should and can be designed as well as how to properly deploy a system. We realize that not all will read this in its entirety, especially not you, but the few things that would be most beneficial of a take away are:

1. Best Practices (found on page 38)
2. understanding the differences unicast (Dante Default) and multicast (Dante Advanced)
3. Maximizing Zones (found on page 27)

IP Enabled Audio Distribution

with Dante® Audio Networking

Nice Home Management IP amps use Dante audio networking technology, enabling standard IP networks to transmit high-quality, uncompressed audio with near-zero latency. Dante is the audio networking choice of nearly all professional audio manufacturers, making it the industry standard for digital audio networking.

Dante®

Dante Audio Networking

- Distributes audio long distances over the network
- Delivers high quality audio with near-zero latency
- Provides lower total cost of ownership than alternative solutions by dramatically reducing cabling and labor costs
- Installs easily using off-the-shelf IT networking equipment and cables, and leverages proven, reliable protocols
- Enables decentralized installations, with sources and amplifiers placed where they make the most sense
- A single cable routes audio
- Standard Networks
- Easily expands over time
- Reliable Audio Network Industry Standard

Designed to Meet Your Needs in Any Room

	Multi-Zone Audio Matrix EL-IPD-MTX-8CH	Multi-Zone Audio Extender EL-IPD-AMP-EXT	Multi-Zone Preamp Audio Matrix EL-IPD-PRE-MSI
			
MODES	Single-Chassis or Multi-Chassis*	Multi-Chassis Only	Single-Chassis or Multi-Chassis*
OUTPUTS	8 Speaker / 4 Preamp	8 Speaker / 4 Preamp	12 Preamp
INPUTS	8 RCA / 1 3.5 / 4 digital	Dante Only	16 RCA / 1 3.5 / 6 digital
SENDING STREAMS *Unicast - each source is streamed separately to each requesting chassis. *MultiCast – each source is counted as a single stream, no matter how many chassis it is sent to.	Unicast (default) = 6 MultiCast = 12 * These are accurate for stereo streams/sources.	N/A	Unicast (default) = 8 MultiCast = 16 * These are accurate for stereo streams/sources.
OUTPUTS ZONES	Up to 12	Up to 8	Up to 12

	Local Audio Source EL-IPD-PRE-SSI	Network Audio Card EL-IPD-NAC-EXT	EL-IPD-AMP-2CH	EL-IPD-PRE-SIO
				
MODES	Multi-Chassis Only	Adds Dante Audio Networking	Single-Chassis or Multi-Chassis	Multi-Chassis Only
OUTPUTS	Dante Only	N/A	1 Zone [Speaker level out (bridgeable) + sub out]	1 zone, Stereo analog out + sub out
INPUTS	RCA / Optical/ Coaxial**	N/A	Single-Chassis: RCA/Optical/Coaxial Multi-Chassis: RCA/Optical/Coaxial**	RCA / Optical/ Co-axial**
SENDING STREAMS	Unicast (default) = 2 MultiCast = 4	N/A	Unicast (default) = 2 Multicast = 4	Unicast (default) = 2 Multicast = 4

*These are accurate for stereo streams/sources

**One Input, that can be set to the connection type

Multi Zone Audio Matrix - Deep Dive

Multi Zone Audio Matrix
The Multi Zone Audio Matrix can be used in a single chassis configuration or with other amplifiers and pre-amps within the Nice Home Management IP audio distribution family of products by adding the optional Dante Audio Networking Card.

EL-IPD-MTX-8CH - Single-Chassis
In single chassis mode, the Multi Zone Audio Matrix supports up to 13 input sources (4 digital, 8 mono / 4 stereo analog RCA and 1 stereo 3.5mm typically used as the doorbell input from an Nice Home Management System Controller). Each RCA analog input can be configured as left, right, or mono 8.

Speaker channels are available (up to 4 stereo zones or 8 mono zones). 4 line level outputs can be configured as 4 mono or 2 stereo output zones or as sub-woofer out. Each can be configured as left, right, mono, or sub out. In Single Chassis mode, the Multi Zone Audio Matrix supports up to 12 individual audio zones.

EL-IPD-MTX-8CH - Dante
Adding the Dante Network Audio Card (EL-IPD-NAC-EXT) allows for additional Nice Home Management AVoIP devices to be used to add additional sources or zones to the system.

EL-IPD-AMP-EXT - Multi Zone Audio Extender

The EL-IPD-AMP-EXT Multi Zone Audio Extender is designed to be used or with other amplifiers and pre-amps within the Nice Home Management IP audio distribution family of products using Dante Audio Networking. The EL-IPD-AMP-EXT Multi Zone Audio Extender is Dante enabled.

NOTE

- The EL-IPD-AMP-EXT Multi Zone Audio Extender cannot be used as a standalone unit.
- It requires another Dante enabled Nice Home Management amplifier (EL-IPD-8CH-MTX Multi Zone Audio Matrix with the EL-IPD-NAC-EXT Dante Network Audio Card) or Preamp (EL-IPD-PRE-MSI Multi Zone Preamp Audio Matrix with the EL-IPD-NAC-EXT Dante Network Audio Card, or EL-IPD-PRE-SSI).

EL-IPD-PRE-MSI - Multi Zone Preamp Audio Matrix

The EL-IPD-PRE-MSI Multi Zone PREAMP Audio Matrix can be used in a stand-alone configuration or with other amplifiers and pre-amps within the Nice Home Management IP audio distribution family of products by adding the optional Dante Audio Networking Card.

EL-IPD-PRE-MSI - Standalone

The EL-IPD-PRE-MSI can be used in stand alone mode paired with external amplifiers to support up to 22 input sources (6 digital, 16 mono / 8 stereo analog RCA and 1 stereo 3.5mm typically used as the doorbell input from an Nice Home Management System Controller). Each RCA analog input can be configured as left, right or as mono. 12 line level outputs can be configured as 12 mono or 6 stereo output zones or as subwoofer out. Each can be configured as left, right, mono, or sub out.

EL-IPD-PRE-MSI - Dante

Adding the Dante Network Audio Card (EL-IPD-NAC-EXT) allows for additional chassis to be used to add additional sources or zones to the system. Combine with the Multi Zone Audio Matrix to add additional sources, or use the EL-IPD-PRE-MSI for sources and add additional EL-IPD-AMP-EXT for audio zones.

EL-IPD-AMP-2CH

The AMP-2CH can be used in Single or Multi-chassis mode. In Single-Chassis mode, each input is independent and can be switched from the UI but cannot send or receive streams from other Dante enabled devices. In Multi-Chassis mode, this device has one input but can send and receive streams to other Dante devices simultaneously on the network. This AMP will supply one zone of output with a Left, Right speaker output (bridgeable) and a sub-out (Variable only; 100hz crossover).

EL-IPD-PRE-SIO

The EL-IPD-PRE-SIO provides a way to deliver two channel PCM audio from a single input of RCA, Optical, or Coax, and then send it to other Dante enabled IPD units on the network. The SIO also adds one zone output consisting of left/right and sub pre-amp outputs. This allows the option to add another zone remotely and create a distributed audio system. The SIO can send and receive audio simultaneously.

EL-IPD-PRE-SSI - Local Audio Source

The EL-IPD-PRE-SSI Local Audio Source provides a way to deliver 2 channel PCM audio from a single remote source to an Nice Home Management Dante enabled Audio Matrix over an existing standard local area network. The EL-IPD-PRE-SSI adds an additional source to the Nice Home Management Dante enabled Audio Matrix.

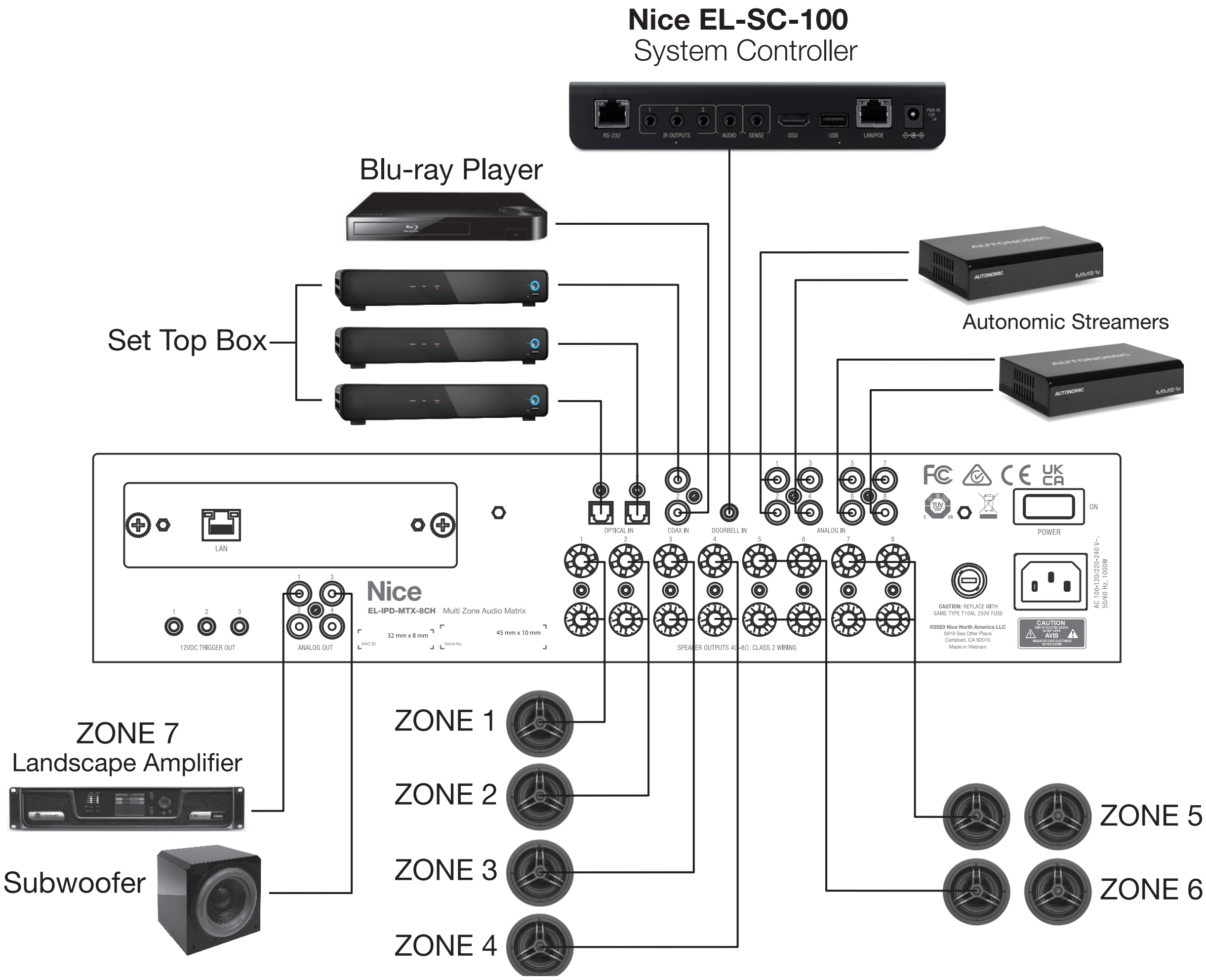
EL-IPD-NAC-EXT - Dante Network Audio Card

The EL-IPD-NAC-EXT enables the Nice Home Management IP amplifiers and pre-amplifiers to be linked in order to share sources and/or audio zones.

The EL-IPD-NAC-EXT Network Audio Card is compatible with the EL-IPD-MTX-8CH Multi Zone Audio Matrix and the EL-IPD-PRE-MSI Multi Zone Preamp Audio Matrix.

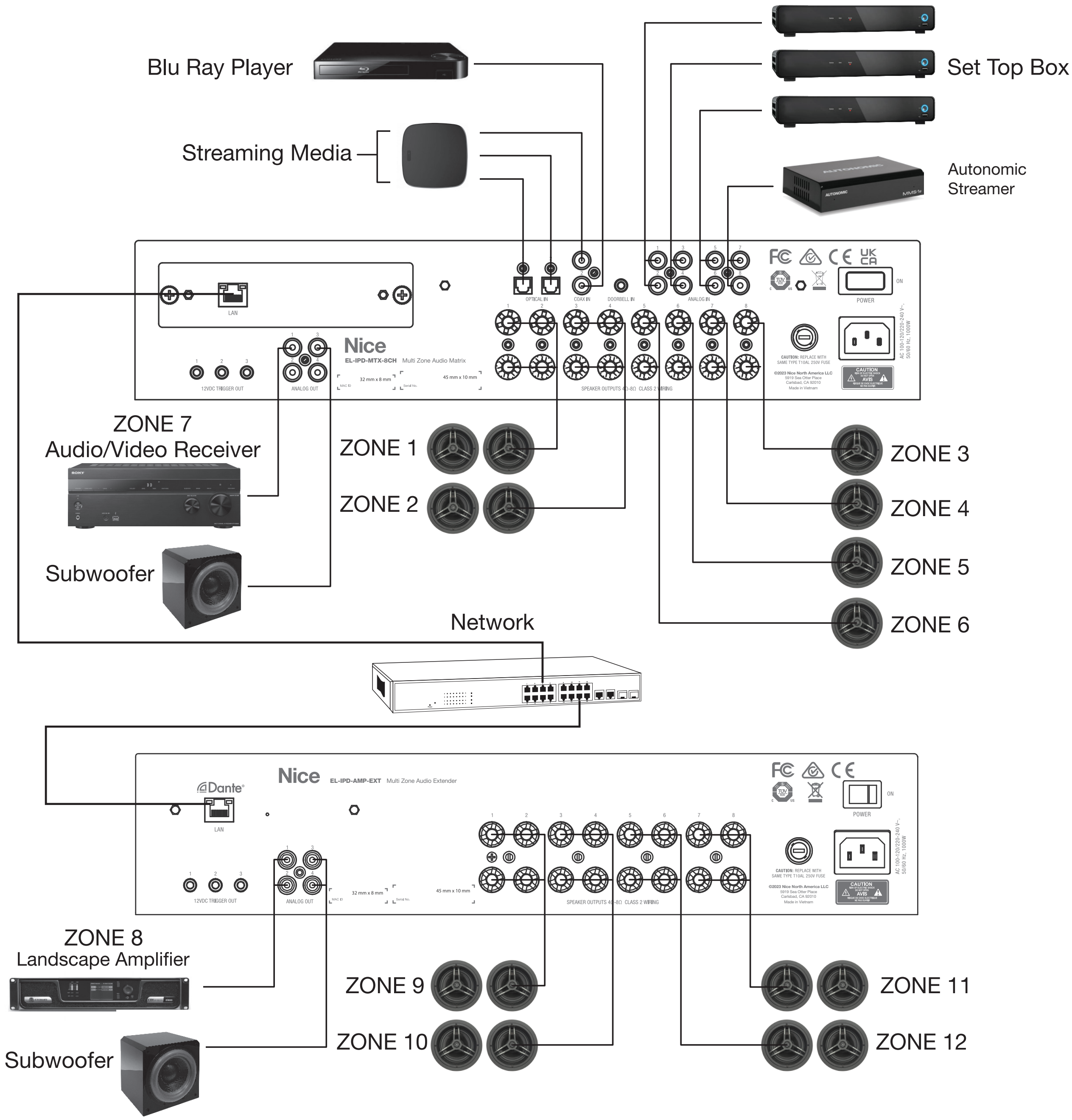
EL-IPD-MTX-8CH - Multi Zone Audio Matrix in standalone mode

The EL-IPD-8CH-MTX can be used in stand alone mode to support up to 13 input sources (4 digital, 8 mono / 4 stereo analog RCA and 1 stereo 3.5mm typically used as the doorbell input from an Nice Home Management System Controller). Each RCA analog input can be configured with another as a L/R pair or individually as mono 8 speaker channels are available (up to 4 stereo zones or 8 mono zones). 4 line level outputs can be configured as 4 mono or 2 stereo output zones or as sub-woofer out. Each can be configured as left, right, mono, or sub out.



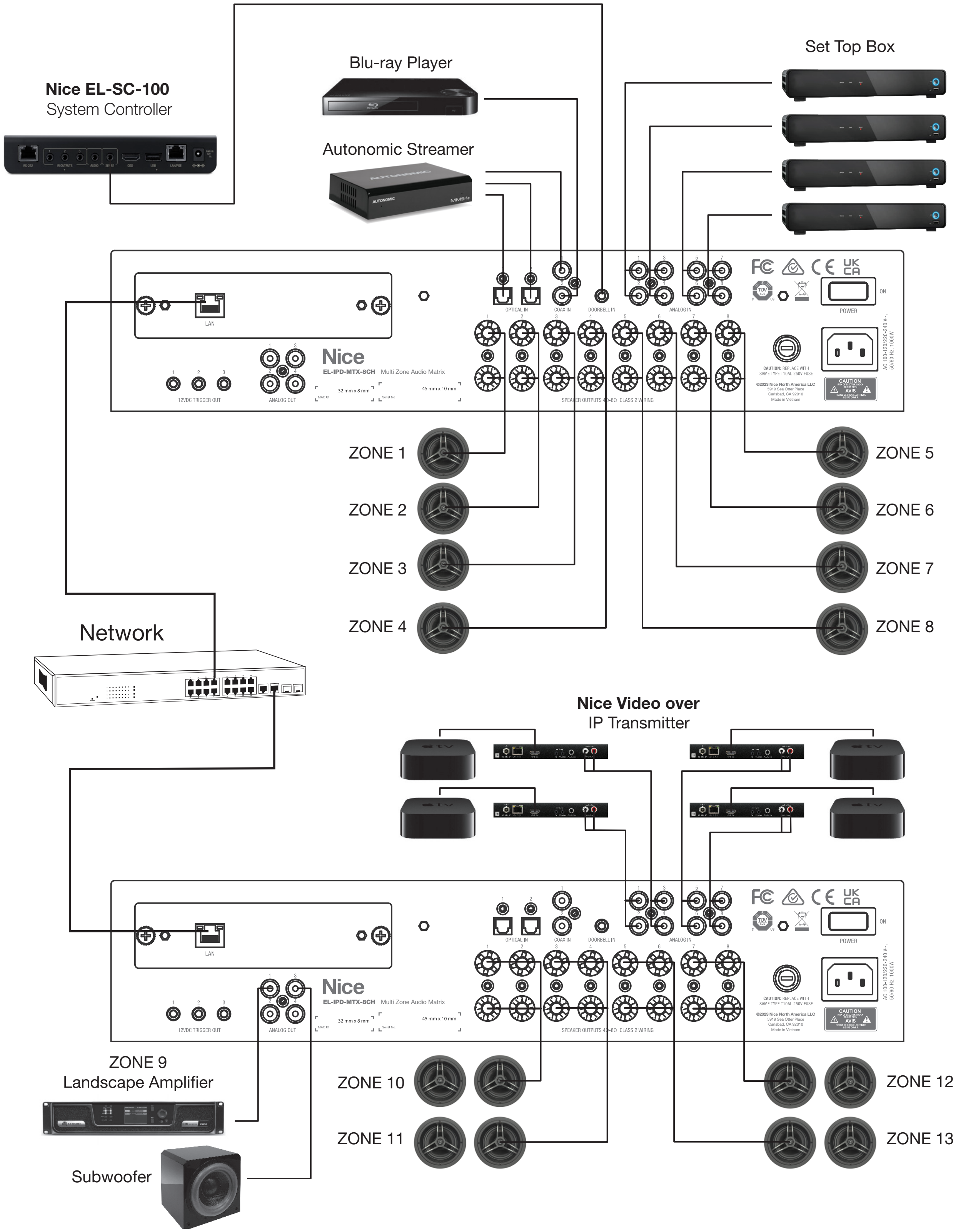
EL-IPD-MTX-8CH - Multi Zone Audio Matrix - EL-IPD-AMP-EXT Multi Zone Audio Extender in Dante Mode

Adding the EL-IPD-NAC-EXT -Network Audio Card gives the ability to combine multiple chassis in to one system. Combine the EL-IPD-MTX-8CH Multi Zone Audio Matrix with the EL-IPD-AMP-EXT Multi Zone Audio Extender to add up to 8 more zones of audio. Using Dante Audio Networking, the two chassis can be linked using standard networking equipment.



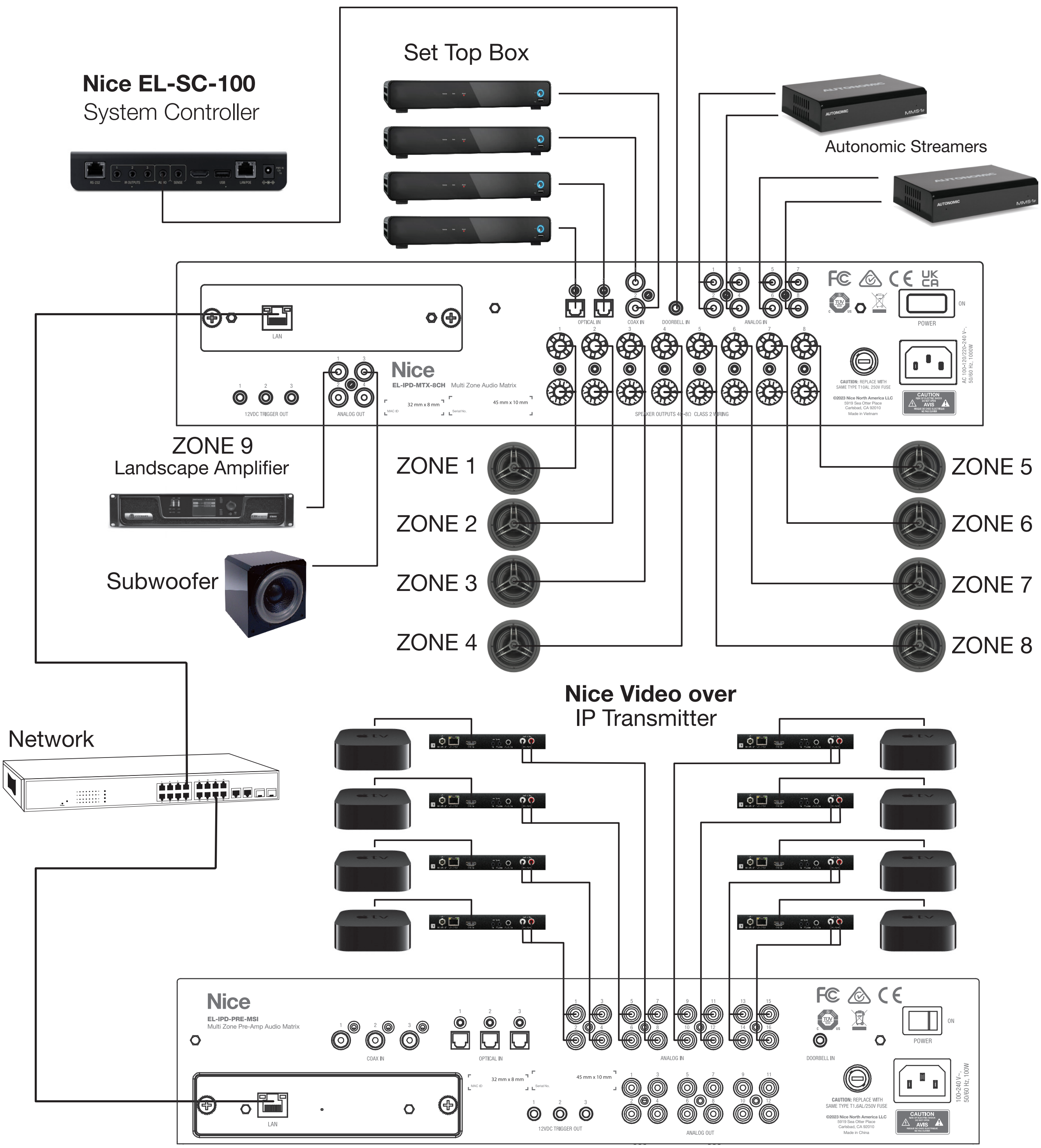
EL-IPD-MTX-8CH - Multi Zone Audio Matrix in Dante Mode - Multiple Chassis

If more zones AND sources are needed beyond the capability of a single EL-IPD-MTX-8CH Multi Zone Audio Matrix, a second can be added to double the number of inputs and zones. By adding the EL-IPD-NAC-EXT - Network Audio Card to both EL-IPD-MTX-8CH Multi Zone Audio Matrix chassis, they can be connected to add up to 12 more zones and 12 inputs. Multiple chassis can be added to create up to 64 zones of audio.



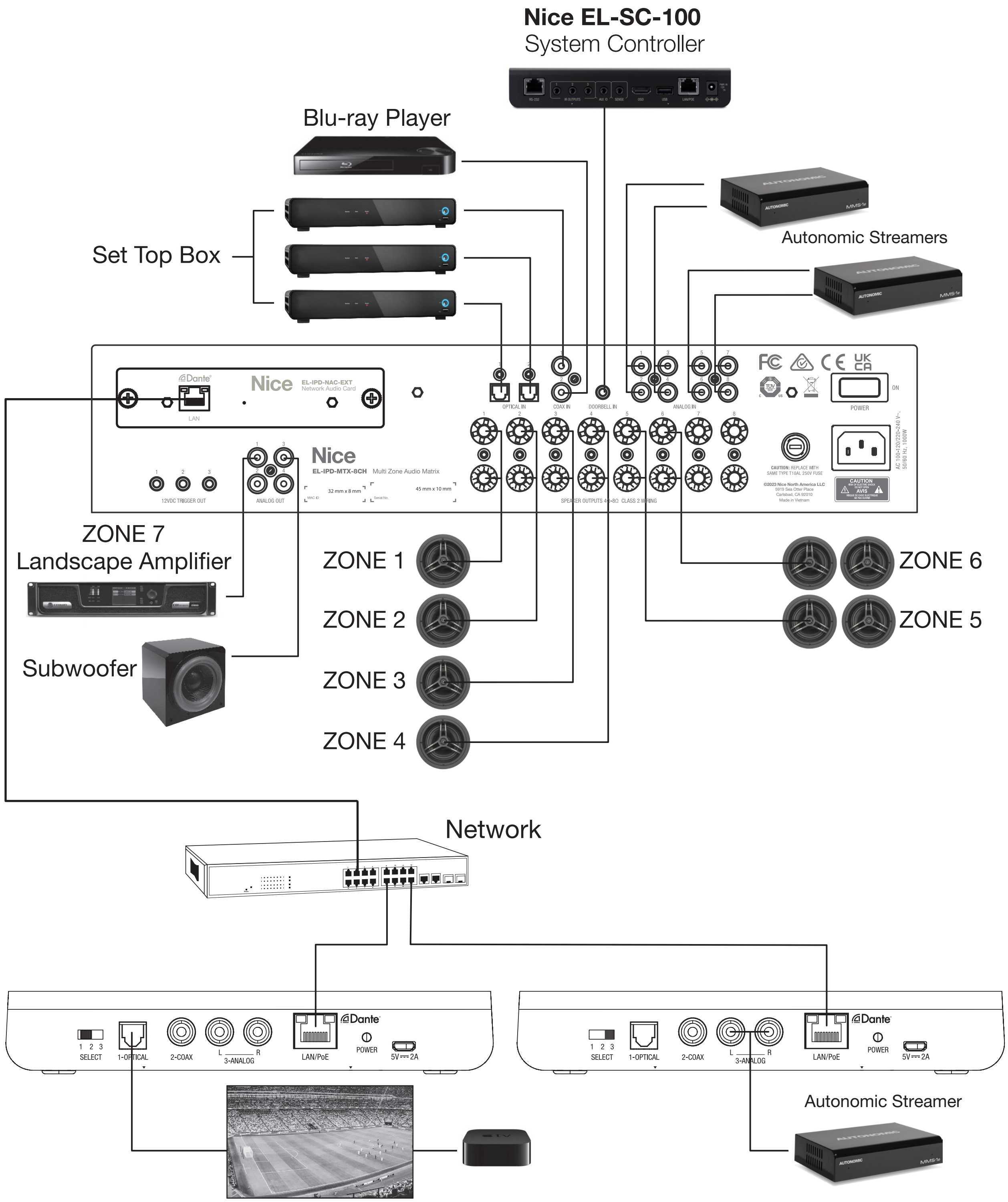
EL-IPD-MTX-8CH - Multi Zone Audio Matrix - EL-IPD-PRE-MSI Multi Zone Preamp Audio Matrix in Dante Mode

If more sources are needed beyond the capability of a single EL-IPD-MTX-8CH Multi Zone Audio Matrix, the EL-IPD-PRE-MSI can be added to increase the number of inputs available. By adding the EL-IPD-NAC-EXT - Network Audio Card to both EL-IPD-MTX-8CH Multi Zone Audio Matrix and the EL-IPD-PRE-MSI Multi Zone Preamp Audio Matrix, they can be connected to add up to 12 preamp zones and 22 inputs. Using Dante Audio Networking, the two chassis can be linked using standard networking equipment. All sources and zones are shared as one system.



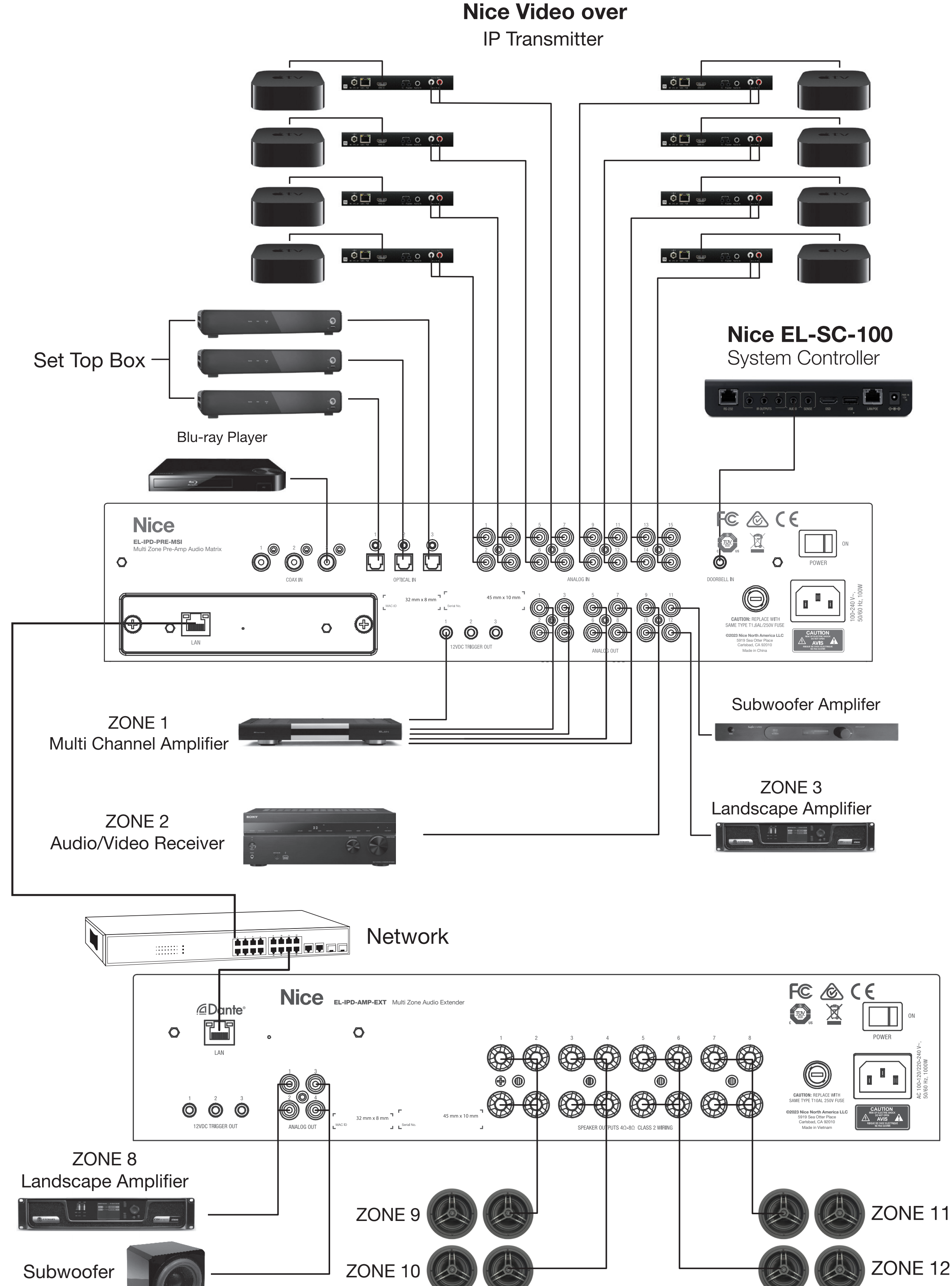
EL-IPD-MTX-8CH - Multi Zone Audio Matrix - EL-IPD-PRE-SSI Local Audio Source in Dante Mode

If remote sources are needed such as a TV's audio output, the EL-IPD-PRE-SSI Local Audio Source input device can be used to send audio over the network to the EL-IPD-MTX-8CH Multi Zone Audio Matrix. Using Dante Audio Networking, the two devices can be linked using standard networking equipment. The EL-IPD-PRE-SSI is an additional source added to the direct input sources on the EL- IPD-MTX-8CH.



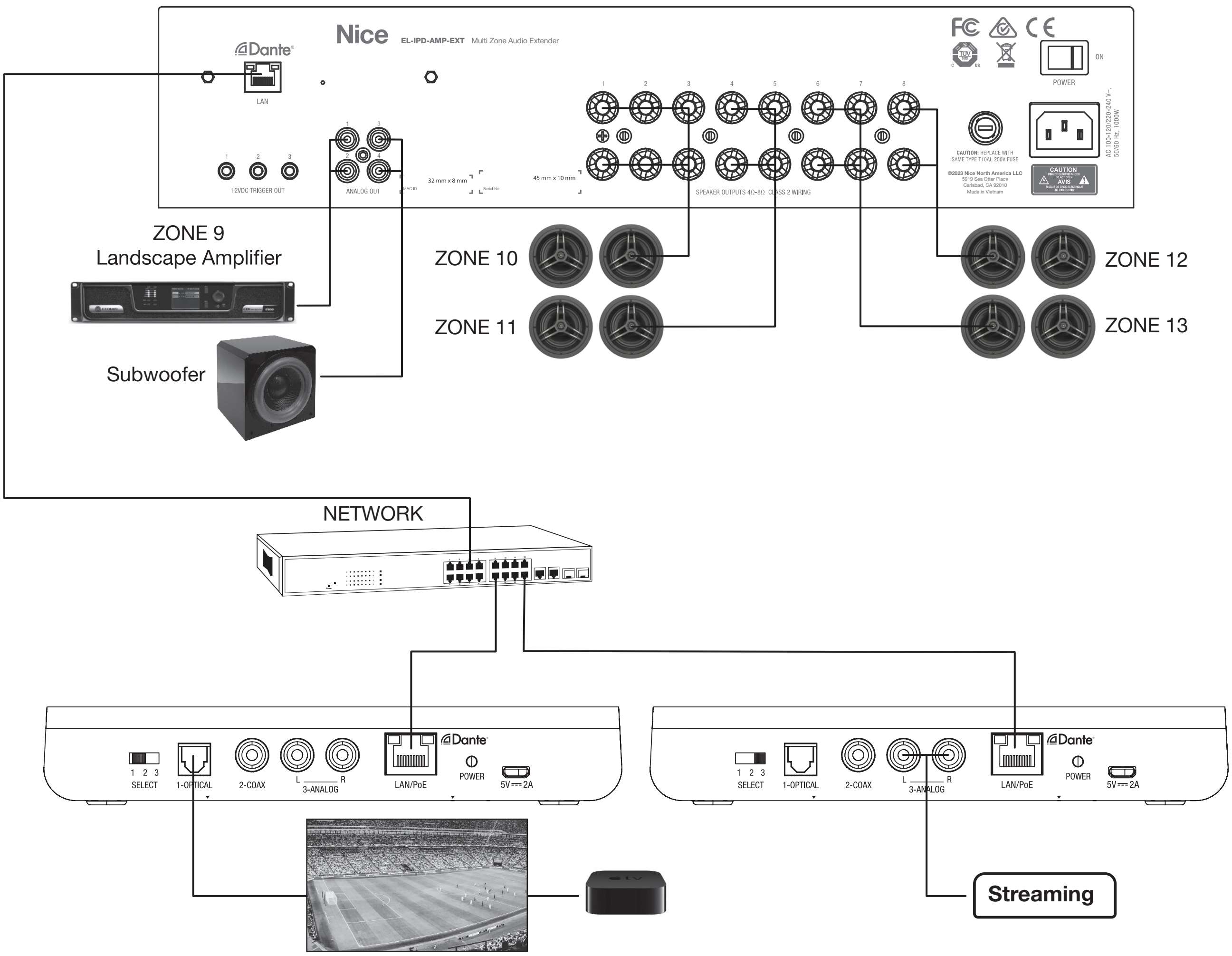
EL-IPD-AMP-EXT - Multi Zone Audio Extender - EL-IPD-PRE-MSI Multi Zone Preamp Audio Matrix in Dante Mode

If more sources are needed beyond the capability of the EL-IPD-MTX-8CH Multi Zone Audio Matrix, the EL-IPD-PRE-MSI can be used for the inputs. With the optional EL-IPD-NAC-EXT - Network Audio Card to added to the EL-IPD-PRE-MSI Multi Zone Preamp Audio Matrix, it can be linked to the EL- IPD-AMP-EXT Multi Zone Audio Matrix.



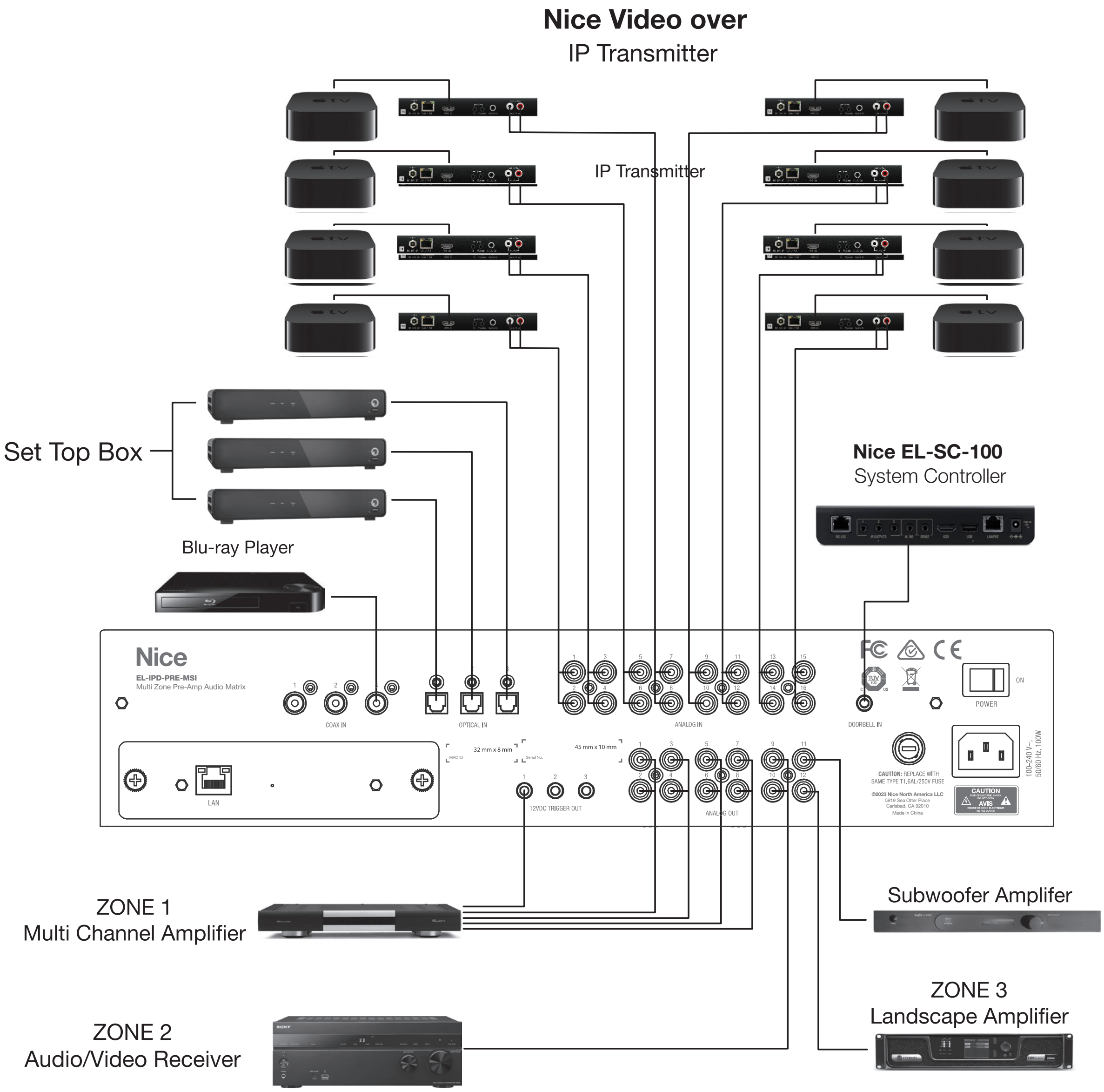
EL-IPD-AMP-EXT - Multi Zone Audio Extender - EL-IPD-PRE-SSI Local Audio Source in Dante Mode

If only remote sources are needed such as from a TV's audio output, the EL-IPD-PRE-SSI Local Audio Source input device can be used to send audio over the network to the EL-IPD-AMP-EXT Multi Zone Audio Extender. Using Dante Audio Networking, the devices can be linked using standard networking equipment.



EL-IPD-PRE-MSI - Multi Zone Preamp Audio Matrix in standalone mode

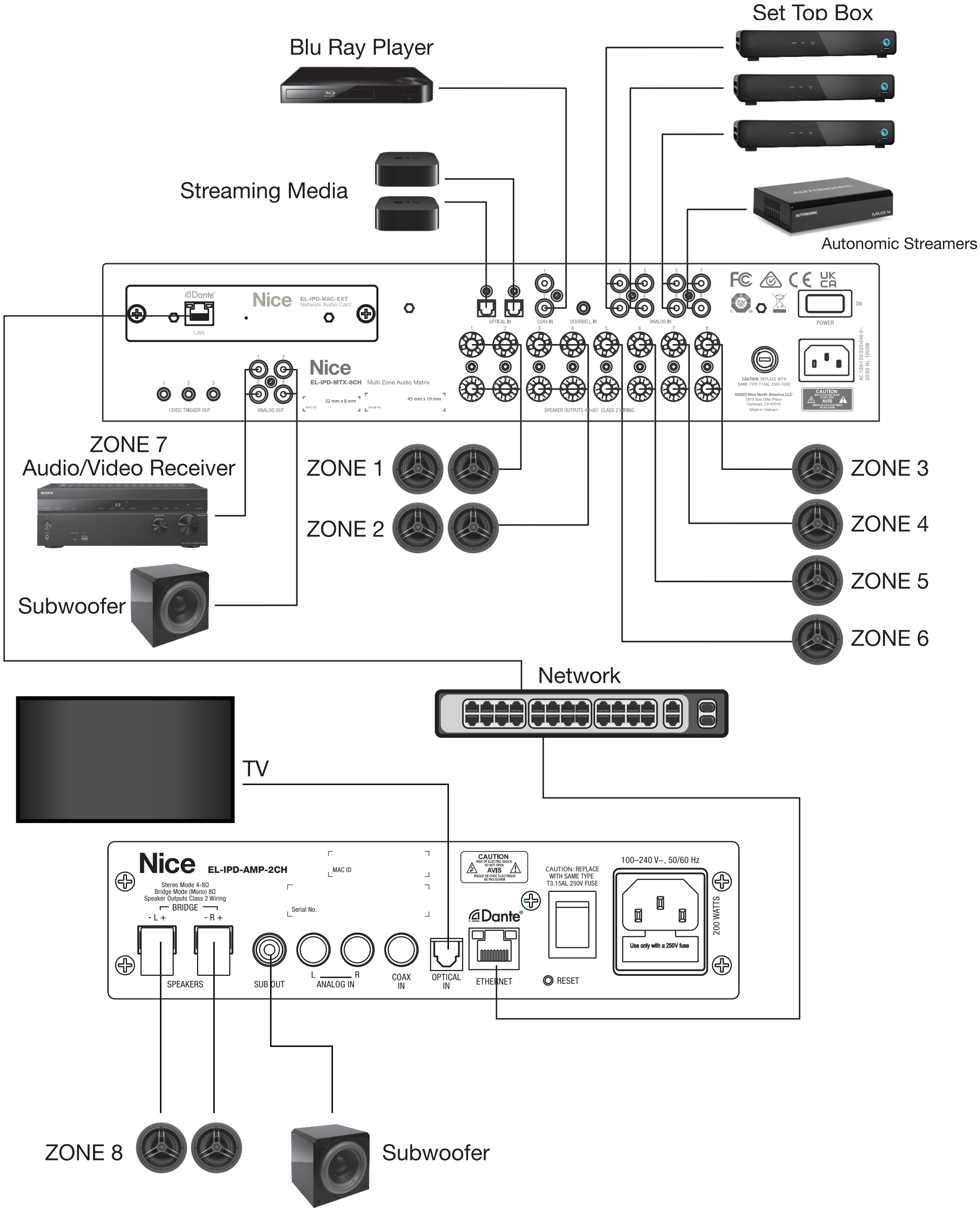
The EL-IPD-PRE-MSI can be used in stand alone mode paired with external amplifiers to support up to 22 input sources (6 digital, 16 mono / 8 stereo analog RCA and 1 stereo 3.5mm typically used as the doorbell input from an Nice Home Management System Controller). Each RCA analog input can be configured as left, right or as mono. 12 line level outputs can be configured as 12 mono or 6 stereo output zones or as subwoofer out. Each can be configured as left, right, mono,or sub out.



DO NOT SELECT THIS without the proper network gear and setup
Fully covered in Advanced Multiple Chassis Configuration – Dante Multicast

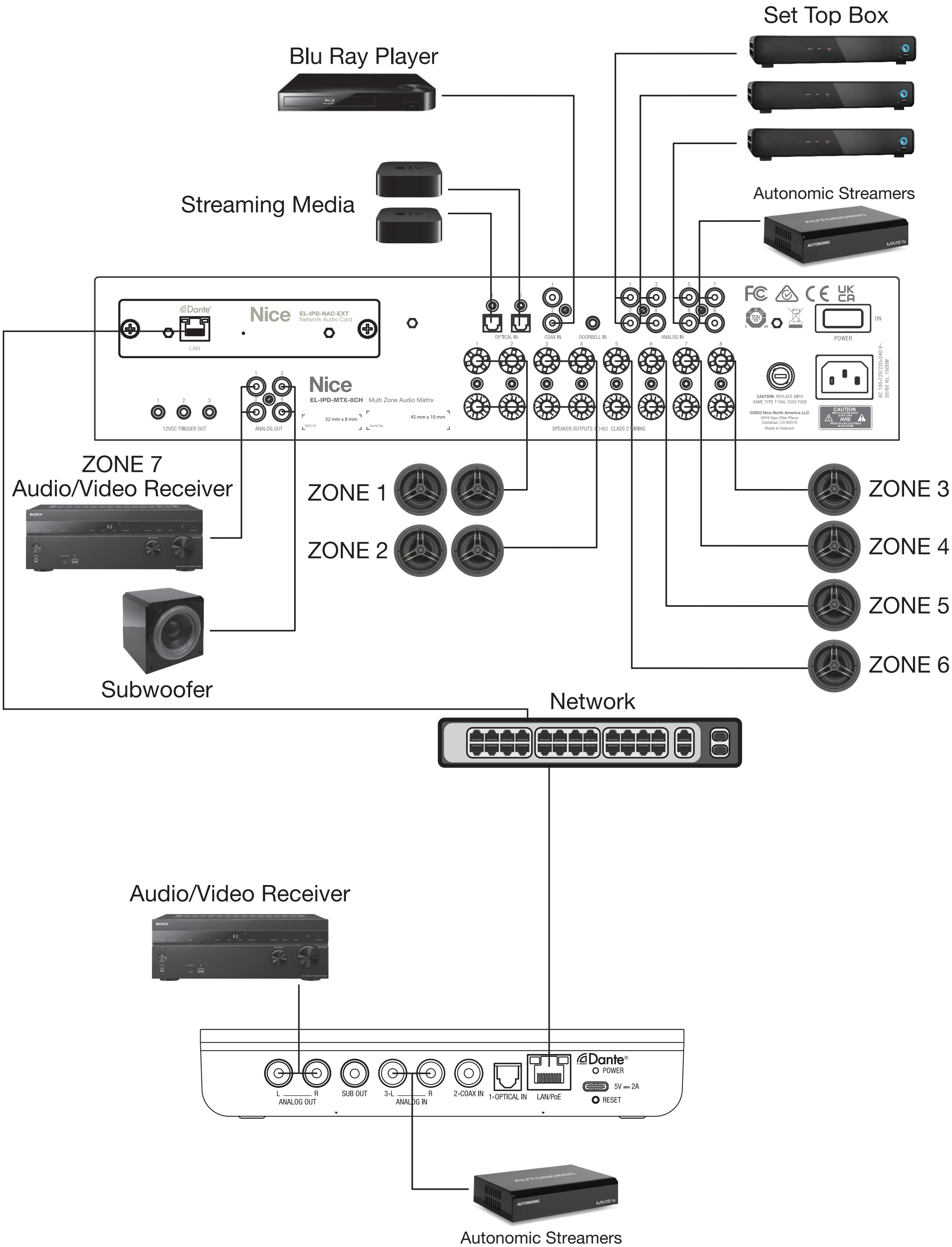
EL-IPD-AMP-2CH - Single-Zone Amplifier in Multi-Chassis Mode

When the AMP-2CH is used in Multi-Chassis mode, it can send and receive a source simultaneously or use the one source (input is a static setting when in Multi-chassis mode). When other IPD devices are part of the system, the sources can be shared across (as seen below).



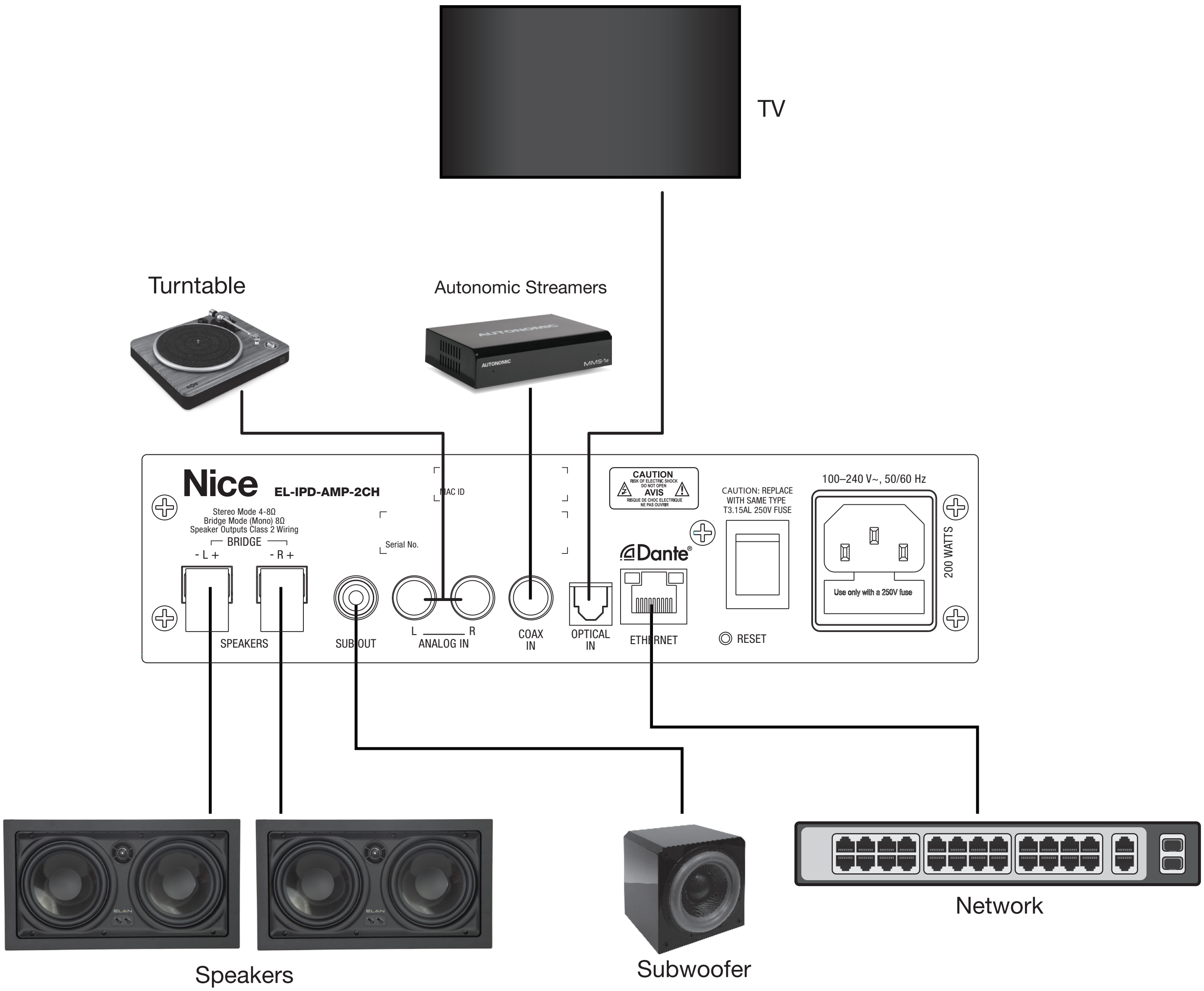
EL-IPD-PRE-SIO - Zone Preamp Output and Source Input

The EL-IPD-PRE-SIO can be used to add one source (input to be set) and/or add one zone output. The zone also has a variable sub output. It can send and receive an audio stream simultaneously.



EL-IPD-AMP-2CH - Single-Zone 2 channel AMP in Stand-alone Mode

When the EL-IPD-AMP-2CH is used in Single-Chassis mode (which it can be set up as, even if other Dante products are in the system), the zone can use and switch between all inputs in real time (like a 3x1 switcher). When used in this mode, it is not possible to send or receive sources from other Dante enabled devices. The zone's speaker outputs are bridgeable, and they can support 8 and 4 ohms. Also featured is a variable sub output.



Functionality Features

The following features are specific to this product line:

Zone Output Configuration

No longer are you held to left/right output per zone or using a dual voice coil in rooms with a single speaker. With the IPD products, you can set a zone to use certain outputs specific to its use. Mono (summed Stereo), Left, Right and Sub channels can be assigned.

Want a room with one speaker or it is desirable to use summed stereo rather than a dedicated left/right? No problem. Don't design your audio zone around the product, design your product around the zone needs.

Audio Ramp

When a zone is either unmuted or powered on, the audio will ramp up, rather than a harsh volume jump. When muted, the volume will decrease to zero immediately.

Mute Rules

If a zone is muted, lowering the volume can be accomplished without unmuting. Then when unmuted, the volume is at the desired level. Rather than unmute at a higher than desired volume, you can lower then unmute.

Raising a volume when a zone is muted will automatically unmute the zone. If you want the audio to be higher, chances are that you desire to hear the audio at that time.

Volume Curve

The volume curve of the IPD series products is specifically tailored to the end user. It offers a non linear curve that allows a lower volume without impacting the high end range.

Equalizer

These products feature a built-in, zone specific EQ. It has six presets, but each preset can be modified and saved. This allows an end user to create and recall a desired curve. Custom curves are set and saved in each zone.

Expandability

There's no need to worry about future system expansion. A source, a zone, or both can be added by adding a SII, SIO or any ELAN IPD audio chassis as the system grows over time (centralized or decentralized).

Zone Specific Volume Parameters

Each zone has three settings for volume that can be adjusted depending on the needs.

Volume Max

Volume Max can be set to limit the volume so that 100% will be a lower volume. When this is set, the volume bar in the user interface will not indicate that this cap has been set. The volume will set to the last known volume, but there are caps on the end that can be adjusted.

Min

Set the zone power up volume Minimum. If the zone powers up with a lower volume than this setting, it will increase the volume to this setting when turned on. This setting is helpful in the event that a zone happened to be set to a volume lower than desired prior to the most recent power down.

Max

Set the zone power up volume Maximum. This setting can be adjusted so that a zone volume will not be higher than this setting after power up. This setting is helpful in the event that a zone happened to be set to a volume higher than desired prior to the most recent power down.

Single vs multi chassis

The EL-IPD-MTX-8CH and EL-IPD-PRE-MSI chassis in the Nice IP Amp family can be used in either single chassis mode or multi chassis mode. The EL-IPD-AMP-EXT and EL-IP-PRE-SSI can only operate in Multi-chassis mode.

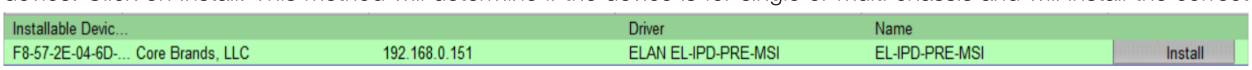
Single chassis mode
Uses the default network card that comes with the unit. With this configuration, the chassis is operating stand-alone – meaning it cannot send or receive audio streams from other devices. It must still be controlled via Nice.

Multi-chassis mode
To use multi chassis mode with the EL-IPD-MTX-8CH and EL-IPD-PRE-MSI chassis, you must use the EL-IPD-NAC-EXT card. Before doing so, you should make sure the chassis is on the latest firmware. In multi chassis mode, the unit can send and receive audio streams. Proper network configuration, limits, and other info found later on.

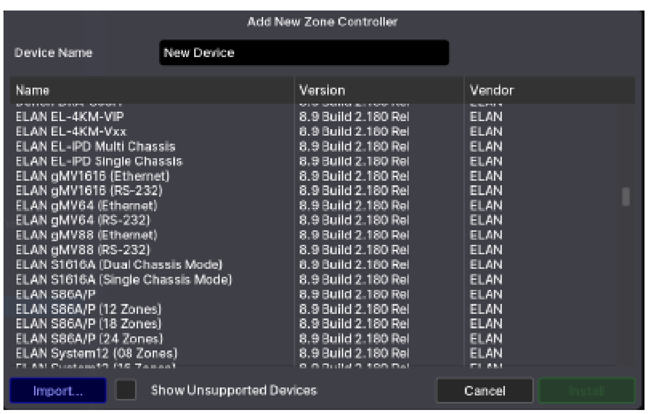
Adding Chassis' to the Configurator

Once you have the chassis connected to the network and powered on, we can now bring it into the configurator. This can be done in 2 different ways. If at some time in the future, another chassis is added, this can be imported in the same way.

Discovery(AD/AC)
You can use the discovery tab to add an Nice IP Amp, either single or multi chassis. In configurator, go to discovery and locate the device. Click on Install. This method will determine if the device is for single or multi chassis and will install the correct driver.

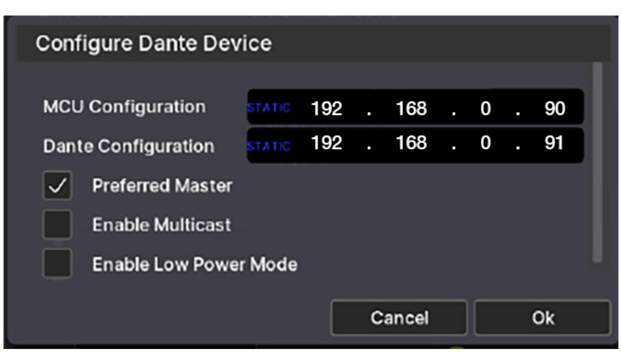


Manually
Unlike the above method, you will have to select single or multi chassis. Go to Media, Zone controllers, and select the correct driver – either Single chassis if using the default network card(and thus are not sending or receiving streams) or Multi-chassis if equipped with a EL-IPD-NAC-EXT card.



Configuring IPD devices

After the chassis' have been added into configurator, we now have to configure some basic settings.



Setting a static IP
Each chassis should be set to a static IP for reliabel opration. If using a NAC-EXT, there will be two IP addresses, one for the MCU (which controls the chassis) and one for Dante (for audio distribution) . Do not confuse this with a DHCP reservation!

Updating firmware
The Nice Home Management IP Amp may need a firmware update. This process can take up to ~15 minutes, during which it should not be powered off for any reason. Any zones configured should be off and stay off while in this process. If an EL-IPD-MTX-8CH or EL-IPD-PRE-MSI, be sure to use the Network card that came with the unit – not the Dante card. If the unit is already installed as multi-chassis, power down, remove EL-IPD-NAC-EXT card, install factory NIC, add as single chassis before updating. After updating, you can power down, and re-install the Dante card. The multi-chassis driver should not have been removed in the process and as such, the chassis upon power up, will now list the current MCU/DSP. You can now remove the single chassis driver, if added.

Device name
Be sure to set a "friendly" name for the chassis when using multi-chassis. This name will be used in the device tree as well as the zone configuration page. It is especially helpful the more chassis that are within the configuration.

Low Power Mode
The Nice Home Management IP Amp chassis are shipped with low power mode on. Upon adding to the configurator, this is set to off as the most common usage is for performance. When in Low Power Mode, the chassis must wake up before executing any command. The wake up time is ~ 20 seconds. Doesn't sound like a lot, but when Sisco comes on, you don't want to wait 20 seconds to hear it! If you are using doorbell, Low Power mode must be off. If on, the doorbell will be over by the time the zones turns on, seemingly like it is not functioning.

Preferred Master
This setting will let you set which chassis will be used for time syncing.

Front LED's Global Latency
Sets the time allowed between chassis. If the audio is delayed more than this amount due to network congestion, audio will not be allowed to pass (and play delayed).

The Nice Home Management IP Amp chassis' have 2 Front Panel LED's. The Below chart is what the different color combination mean:

Condition	Amplifier Status	Power/Network Status
Power On, Booting	Off	Flashing Blue
Power On, Connected to Network	Off	Solid Blue
Self-Check Failure	Off	Flashing Red
Boot Failure	Solid Red	Solid Red
No Network Found	Off	Solid Amber
Power Off	Off	Off
Chassis in Low Power Mode	Solid Blue	Solid Blue
Chassis in Sleep Mode	Off	Solid Red
Amplifier or Preamplifier Channel Clipping	Solid Amber	Solid Blue
Amplifier Channel in Protection Mode	Solid Red	Solid Blue
Chassis Identify Mode	Flashing Amber	Flashing Blue

LED Flashing
The option to flash the LED is for unit identification. If you have multiple chassis, you may want to identify a specific one, if not labeled. To accomplish this, you can select LED flash on and the two front LED's will begin to blink. Once you have identified the chassis you are looking for, you can then selected LED flash stop.

MultiCAST Option (EL-IPD-MTX-8CH, EL-IPD-PRE-MSI, EL-IP-PRE-SSI, EL-IPD-PRE-SIO, EL-IPD-AMP-2CH)
This option is found only on multi chassis systems, and chassis EL-IPD that can send sources. It is for larger systems or systems in which the network residing has IGMP on. Dante can send audio in one of two ways.

Unicast Default (Multicast Unchecked)
By default we use Unicast to send audio streams. Unicast is simpler to deploy as it requires basic networking gear and setup/connections but has its limits as far as how many sources can be streamed.

Multicast
The multicast option changes how the chassis deliver, or send, a source stream. Sending via multicast increases the streaming limits as noted in the chassis overview, but REQUIRES specific networking gear as well as proper networking setup and planning.

DO NOT SELECT THIS without the proper network gear and setup
Fully covered in Advanced Multiple Chassis Configuration – Dante Multicast

EL-IP-PRE-SSI/SIO LED Brightness
This option is only found on the EL-IP-PRE-SSI device. If the device will be behind a TV in a bedroom, for example, the on board LED can emit light in a dark room and may be advantageous to lower its brightness.

Resources

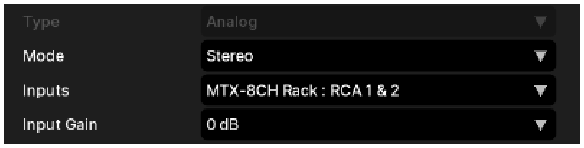
Please reference the Integration Notes, User guides, and videos.

Sources

Adding a Source

We will see how to add an analog as well as a digital source and the options thereof. Upon adding an Nice Home Management IP Amp to the configurator, the available sources will automatically be loaded in, configured for default stereo use.

Analog
Analog sources are added in the default setting of stereo. However, they can be configured for mono use, which would free up an input to be able to add another source.



Stereo (default Mode)
Connect the Left and Right channel of the source to the input. Then assign the source to that input within Configurator.

Mono (optional Mode)
The benefit of using mono(summed stereo) is it allows you to add another source using the orphaned rca input. For example, an EL-IPD-MTX-8CH has 4 analog stereo source inputs. If set to Mono, that can be expanded all the way to 8(all analog sources mono).

Inputs
By default the source assignments come predefined to the appropriate analog inputs in Left/Right pairs. If set to mono, you will have to set the input to RCA. Once that is completed, you will now have an orphaned RCA to add a new mono source location for the orphaned RCA, right click on sources and select Add New Source. Then, assign the new input to the orphaned RCA.

Input level gain
For analog sources, either stereo or mono, set the input gain for the particular source (+/-6 db).

Doorbell
The doorbell input is also analog but it does differ as it is stereo only. It can be used as another source input, and not only for a doorbell input, if you were not using a doorbell.

Digital (Optical/Coaxial)
A digital, coaxial or optical, input can also be used, when available. These are stereo only and do not have an input level adjustment.

EL-IPD-PRE-SSI/SIO (Multi-chassis only)
Another option for an input is the use of a EL-IP-PRE-SSI/SIO which can add an additional source, even ones remotely located. The EL-IP-PRE-SSI/SIO allows for 1 source to be connected, and thus added to the multi-chassis system. It can be either a coaxial, optical or analog (stereo only) connection type. Set the input type switch on the rear of the EL-IP-PRE-SSI to coincide with the connection type used. The EL-IPD-PRE-SIO must have the input selected (Analog, Coax, or Optical) in the Input drop down in configurator. The EL-IP-PRE-SSI/SIO can send its source to a max of 2 separate chassis simultaneously in unicast (default) mode. If there is a need for more than 2 simultaneous chassis, then multicast is needed.

DO NOT SELECT THIS without the proper network gear and setup
Fully covered in Advanced Multiple Chassis Configuration – Dante Multicast

Configuring Zones

Zones come pre-configured for stereo usage out of the box. However with an Nice Home Management IP Amp chassis' the zones can be configured in a multitude of ways making this a unique offering. A zone can consist of a single output (speaker or pre-amp), or multiple outputs (speaker, preamp or any combination of the two). By allowing this level of customization, the Nice Home Management IP Amp can truly be configured in a way that optimizes your audio project. No longer are you bound to a pair of speakers or dual voice coil speakers! (enter highlighting music) That also means the amount of zones per chassis is dependent on the zone configuration. For example, an EL-IPD-MTX-8CH can, in one extreme, have each output set up as an independent zone. The 8 speaker outputs(which are 4 ohm stable and able to handle summed-stereo) and 4 pre-outs add up to 12 independent zones vs the default of 6 zones (4 pairs of speaker outputs and 2 pairs of pre-outputs).

To configure the zones, begin by navigating to the chassis, zones, then Configuration:

Output & Device

Output	Zone	Type	Impedance	Line Level	Gain	Lip Sync Delay
Amp 1	Zone 1	Left	4 Ohms		0 dB	0 msec
Amp 2	Zone 1	Right	4 Ohms		0 dB	0 msec
Amp 3	Zone 2	Left	4 Ohms		0 dB	0 msec
Amp 4	Zone 2	Right	4 Ohms		0 dB	0 msec
Amp 5	Zone 3	Left	4 Ohms		0 dB	0 msec
Amp 6	Zone 3	Right	4 Ohms		0 dB	0 msec
Amp 7	Zone 4	Left	4 Ohms		0 dB	0 msec
Amp 8	Zone 4	Right	4 Ohms		0 dB	0 msec
Pre-amp 1	Zone 5	Left		Variable	0 dB	0 msec
Pre-amp 2	Zone 5	Right		Variable	0 dB	0 msec
Pre-amp 3	Zone 6	Left		Variable	0 dB	0 msec
Pre-amp 4	Zone 6	Right		Variable	0 dB	0 msec

Are read only information. Indicates which chassis and which output(s) you are configuring. You can select multiple rows to configure multiple outputs of the same zone, for example, by using shift/ctrl.

Zone

Output	Zone	Type	Impedance	Line Level	Gain	Lip Sync Delay
Amp 1	Zone 1	Left	4 Ohms		0 dB	0 msec
Amp 2	Zone 1	Right	4 Ohms		0 dB	0 msec
Amp 3	Zone 2	Left	4 Ohms		0 dB	0 msec
Amp 4	Zone 2	Right	4 Ohms		0 dB	0 msec
Amp 5	Zone 3	Left	4 Ohms		0 dB	0 msec
Amp 6	Zone 3	Right	4 Ohms		0 dB	0 msec
Amp 7	Zone 4	Left	4 Ohms		0 dB	0 msec
Amp 8	Zone 4	Right	4 Ohms		0 dB	0 msec
Pre-amp 1	Zone 5	Left		Variable	0 dB	0 msec
Pre-amp 2	Zone 5	Right		Variable	0 dB	0 msec
Pre-amp 3	Zone 6	Left		Variable	0 dB	0 msec
Pre-amp 4	Zone 6	Right		Variable	0 dB	0 msec

This is where you assign a zone to an output, add a new or remain an existing zone. By default, we pre-populate zones in a common stereo format. To change, right click on the output(s) and select one of the options. For the EL-IPD-MTX-8CH and EL-IPD-PRE-MSI, each output can be defined as its own zone, or up to 12 zones. The EL-IPD-AMP-EXT can be configured for up to 8 zones, of any combination.

Type

Use this to set the output type. Options are Left, Right Mono (summed stereo) or Subwoofer.

Output	Zone	Type	Impedance	Line Level	Gain	Lip Sync Delay
Amp 1	Zone 1	Left	4 Ohms		0 dB	0 msec
Amp 2	Zone 1	Right	4 Ohms		0 dB	0 msec
Amp 3	Zone 2	Left	4 Ohms		0 dB	0 msec
Amp 4	Zone 2	Right	4 Ohms		0 dB	0 msec
Amp 5	Zone 3	Left	4 Ohms		0 dB	0 msec
Amp 6	Zone 3	Right	4 Ohms		0 dB	0 msec
Amp 7	Zone 4	Left	4 Ohms		0 dB	0 msec
Amp 8	Zone 4	Right	4 Ohms		0 dB	0 msec
Pre-amp 1	Zone 5	Left		Variable	0 dB	0 msec
Pre-amp 2	Zone 5	Right		Variable	0 dB	0 msec
Pre-amp 3	Zone 6	Left		Variable	0 dB	0 msec
Pre-amp 4	Zone 6	Right		Variable	0 dB	0 msec

Left/Right: to set an output to either the left or right channel in a stereo zone setting

Mono (Blended stereo): Mono has a bad association in this industry that stems from old AM radio. Stereo audio requires a dedicated listening position for your ears to discern the left and right channels which is often lost with distributed audio. But now each area can be designed and set appropriately. In an area with a TV, the couch is typically the primary listening position would be better suited in stereo, but a kitchen with no dedicated listening position, and lots of moving, would better be served in blended stereo. Which could be done on via a single output zone, increasing your zone outputs as well as speakers per output(which equals happier customers and higher profits).

Subwoofer: As seen in the image, an output can be set to subwoofer. This setting changes the crossover for the output specific for subwoofer use @100Hz.

Impedence

Output	Zone	Type	Impedance	Line Level	Gain	Lip Sync Delay
Amp 1	Zone 1	Left	4 Ohms		0 dB	0 msec
Amp 2	Zone 1	Right	4 Ohms		0 dB	0 msec
Amp 3	Zone 2	Left	4 Ohms		0 dB	0 msec
Amp 4	Zone 2	Right	4 Ohms		0 dB	0 msec
Amp 5	Zone 3	Left	4 Ohms		0 dB	0 msec
Amp 6	Zone 3	Right	4 Ohms		0 dB	0 msec
Amp 7	Zone 4	Left	4 Ohms		0 dB	0 msec
Amp 8	Zone 4	Right	4 Ohms		0 dB	0 msec
Pre-amp 1	Zone 5	Left		Variable	0 dB	0 msec
Pre-amp 2	Zone 5	Right		Variable	0 dB	0 msec
Pre-amp 3	Zone 6	Left		Variable	0 dB	0 msec
Pre-amp 4	Zone 6	Right		Variable	0 dB	0 msec

The Nice Home Management IP Amp is a 4ohm stable amplifier. You can confidently connect a 4 ohm load to any number of speaker outputs. By default this option is set to 4, not 8. Its set this way because, even though unlikely, if a 4 ohm speaker/load was connected to the output on 8 ohm, that would render the amp's output higher than the stated 80 WPC(@ 8ohm) and could potentially deliver too much power to the speaker, if its max happen to be 80wpc in 8 ohm. A 4 ohm load will render the output slightly above the rated 80 watts if a 4 ohm load is connected while set to 8 ohms. In the event of the opposite, If you connected an 8 ohm load while it was set to 4, the only negative would be slightly lower output volume but no potential damage to any connected speakers. If you connect a 2 ohm load I will fly to the job and read you the riot act.

Line Level

Output	Zone	Type	Impedance	Line Level	Gain	Lip Sync Delay
Amp 1	Zone 1	Left	4 Ohms		0 dB	0 msec
Amp 2	Zone 1	Right	4 Ohms		0 dB	0 msec
Amp 3	Zone 2	Left	4 Ohms		0 dB	0 msec
Amp 4	Zone 2	Right	4 Ohms		0 dB	0 msec
Amp 5	Zone 3	Left	4 Ohms		0 dB	0 msec
Amp 6	Zone 3	Right	4 Ohms		0 dB	0 msec
Amp 7	Zone 4	Left	4 Ohms		0 dB	0 msec
Amp 8	Zone 4	Right	4 Ohms		0 dB	0 msec
Pre-amp 1	Zone 5	Left		Variable	0 dB	0 msec
Pre-amp 2	Zone 5	Right		Variable	0 dB	0 msec
Pre-amp 3	Zone 6	Left		Variable	0 dB	0 msec
Pre-amp 4	Zone 6	Right		Variable	0 dB	0 msec

Fixed or Variable (Default) Use this to set a Pre-amp's output to fixed or variable

Gain +/-12db

Output	Zone	Type	Impedance	Line Level	Gain	Lip Sync Delay
Amp 1	Zone 1	Left	4 Ohms		0 dB	0 msec
Amp 2	Zone 1	Right	4 Ohms		0 dB	0 msec
Amp 3	Zone 2	Left	4 Ohms		0 dB	0 msec
Amp 4	Zone 2	Right	4 Ohms		0 dB	0 msec
Amp 5	Zone 3	Left	4 Ohms		0 dB	0 msec
Amp 6	Zone 3	Right	4 Ohms		0 dB	0 msec
Amp 7	Zone 4	Left	4 Ohms		0 dB	0 msec
Amp 8	Zone 4	Right	4 Ohms		0 dB	0 msec
Pre-amp 1	Zone 5	Left		Variable	0 dB	0 msec
Pre-amp 2	Zone 5	Right		Variable	0 dB	0 msec
Pre-amp 3	Zone 6	Left		Variable	0 dB	0 msec
Pre-amp 4	Zone 6	Right		Variable	0 dB	0 msec

This is useful for balancing an output to tailor to a room or other outputs.

Lip Sync 0-400ms.

Output	Zone	Type	Impedance	Line Level	Gain	Lip Sync Delay
Amp 1	Zone 1	Left	4 Ohms		0 dB	0 msec
Amp 2	Zone 1	Right	4 Ohms		0 dB	0 msec
Amp 3	Zone 2	Left	4 Ohms		0 dB	0 msec
Amp 4	Zone 2	Right	4 Ohms		0 dB	0 msec
Amp 5	Zone 3	Left	4 Ohms		0 dB	0 msec
Amp 6	Zone 3	Right	4 Ohms		0 dB	0 msec
Amp 7	Zone 4	Left	4 Ohms		0 dB	0 msec
Amp 8	Zone 4	Right	4 Ohms		0 dB	0 msec
Pre-amp 1	Zone 5	Left		Variable	0 dB	0 msec
Pre-amp 2	Zone 5	Right		Variable	0 dB	0 msec
Pre-amp 3	Zone 6	Left		Variable	0 dB	0 msec
Pre-amp 4	Zone 6	Right		Variable	0 dB	0 msec

It is useful for syncing a TV's sound to the video, especially when also sending video over a balun or matrix. It can also be used to sync two zones where a common listening position favors one of the zones and the zones are far enough apart causing a slight echo effect in the listening position. This is more common with outdoor zones.

Maximizing Zones

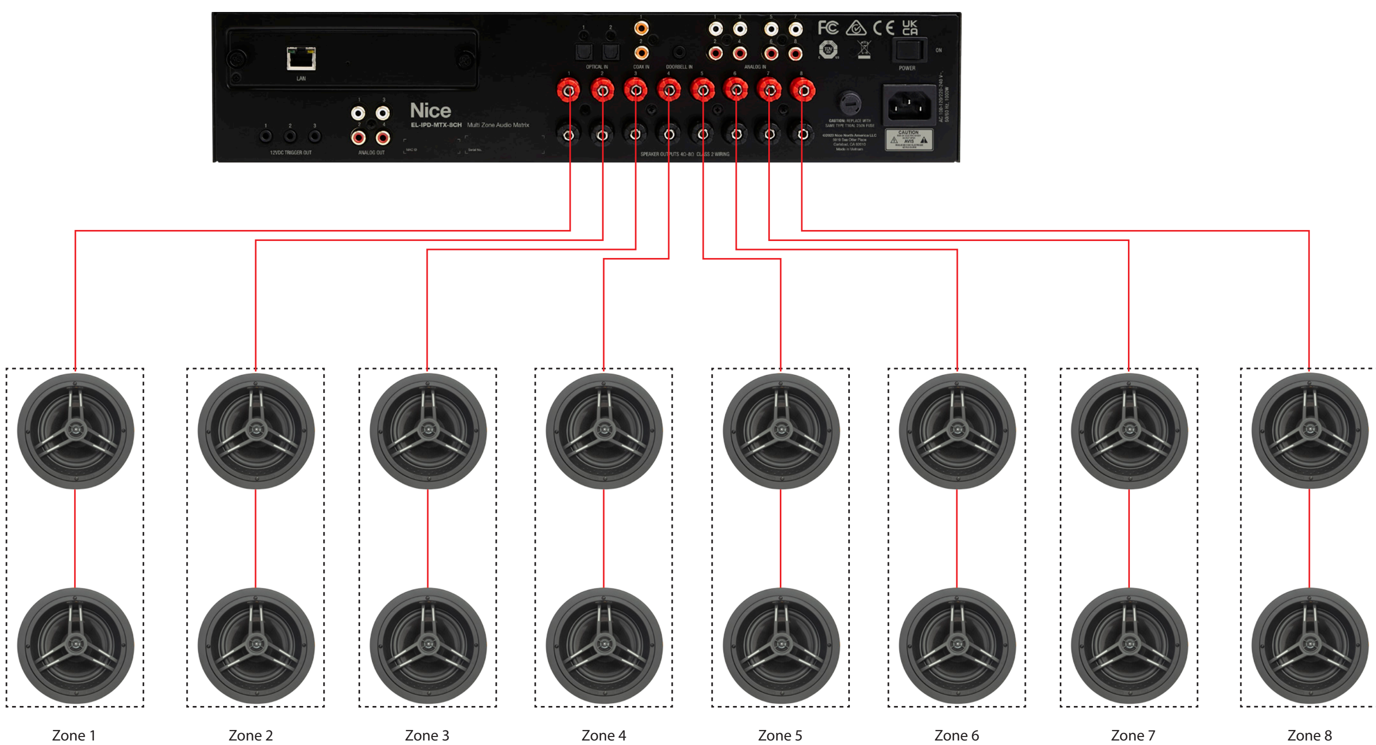
Speaker Configuration

When thinking about the number of zones and speakers needed, it's helpful to keep in mind that a zone is not a speaker and a speaker is not a zone. Zones can be made up of one speaker output, multiple speaker outputs, preamp outputs, or a combination of speaker and preamp outputs. Distributed audio is rarely used where stereo separation is needed - typically other than a media zone where there is a defined listening area. Up to two 8Ω speakers can be connected to each set of speaker terminals for a total of 16 speakers allowing for up to 8 zones.

Tips:

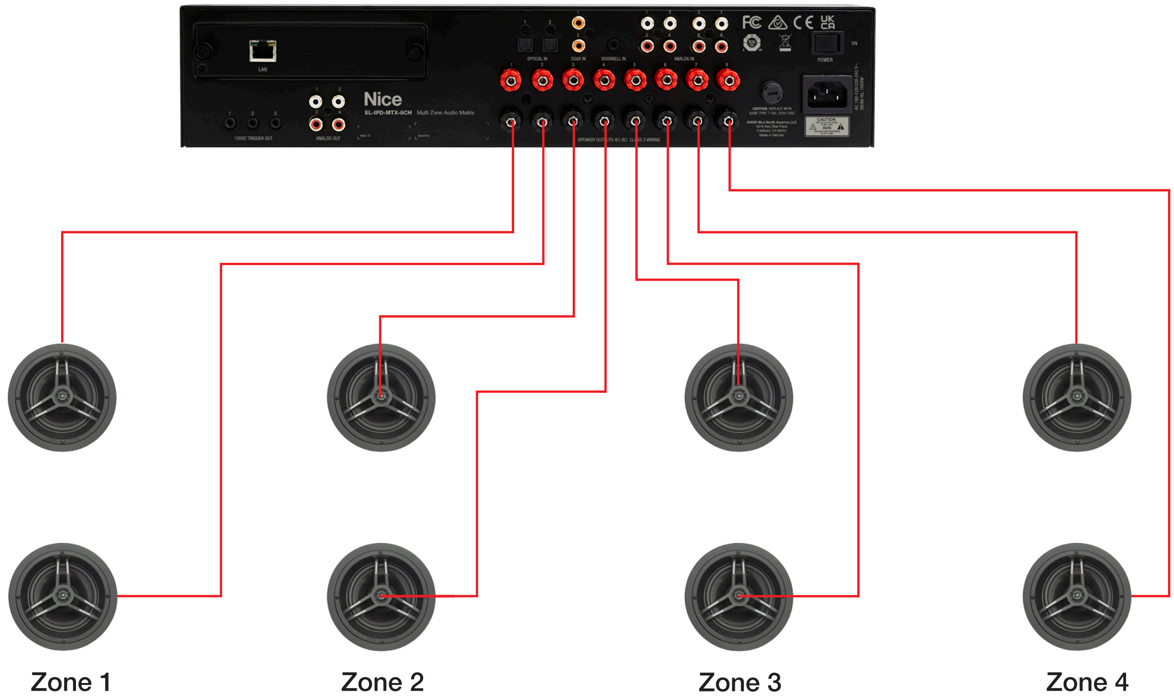
- When connecting two 8Ω speakers to a single speaker output, the impedance will drop to 4Ω. Be sure to set this correctly in Configurator (see section 2 - Configuration Basics).
- To maximize the number of zones, consider using a single speaker in bathrooms or other small rooms or using a pair of speakers on a single speaker terminal. Be sure designate speakers used in summed stereo as mono (see section 2 - Configuration Basics).
- Use a preamp output to add a subwoofer to any zone (see section 2 - Configuration Basics)

Single Output

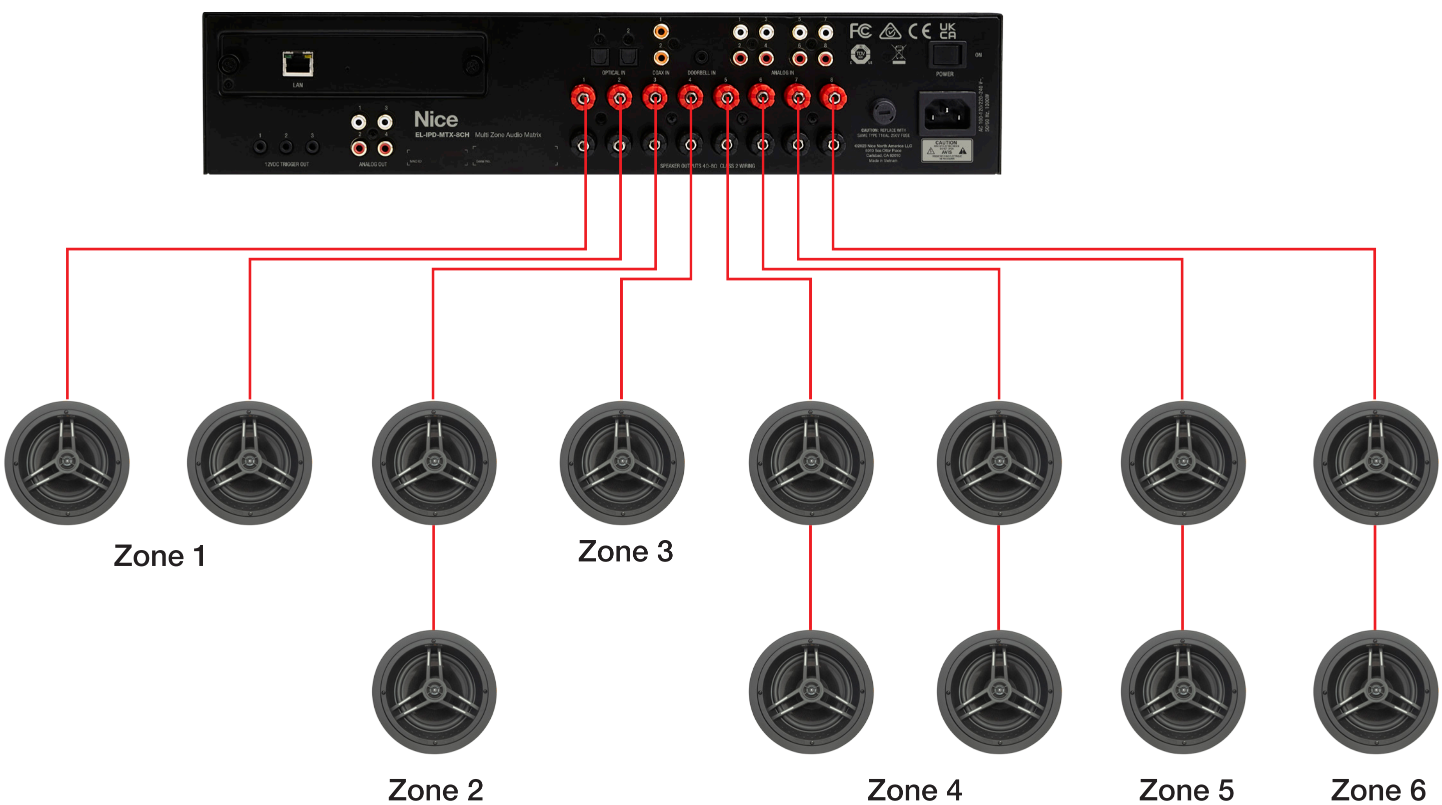


A single output (speaker or Pre-Amp) can be used as a zone and is common in a few use cases. A single output when set to mono will blend the left and right channels together for a blended stereo output. Typical use cases: Kitchen, Outdoor audio, single speaker zone like a bathroom or laundry room, hallways and other common areas with no listening position. No need to use a dual voice coil speaker! Sometimes a zone's main purpose is not for audio believe it or not. One could find the use case of hearing the doorbell or an audio page equally or more desirable in some areas that were not chosen for audio, thus adding more zones and more speakers to a project.

Stereo Output (Default)



Multiple output



Pre Out Configuration

The Nice Home Management IP Amp pre outs are configurable just like the speaker outputs. They can be used in conjunction with a zone that has speaker outputs or can be used as a separate zone! They can be set to a fixed or variable output as well. Lets take a look at some common uses.

Pre-out zone / 70 volt / AVR
If we set the output to its own zone, and have the pre-outs set to variable, we will be able to control the volume as these are feeding an external amplifier. We can also set the output(s) to fixed if we were feeding a volume control. It can also be used as a single output, like the speaker outputs or with two for Stereo or any combination thereof.

Split zones
We can also set up multiple outputs to the same zone in which they can be set to fixed and another set to variable in the same zone for a truly custom scenario, like a zone with volume control and a sub zone where volume is controlled by a volume control.

Sub output
Using a pre-out for a sub is also a common use case. This can be done easily by setting the output to the zone it will be a part of. Typically, you will set this to be a variable output. The gain adjustment is useful for dialing in the bass, after the sub has been positioned.

Multiple Chassis Configuration - Dante default (unicast)

The default Multi chassis operating mode is set to use Unicast streaming. This requires much less network configuration and setup as well as lower equipment requirements but comes with a lower limitation of sending streams. As such, it is stated that this configuration can support an average of 4 chassis. Of course design and application play a large part.

Unicast overview

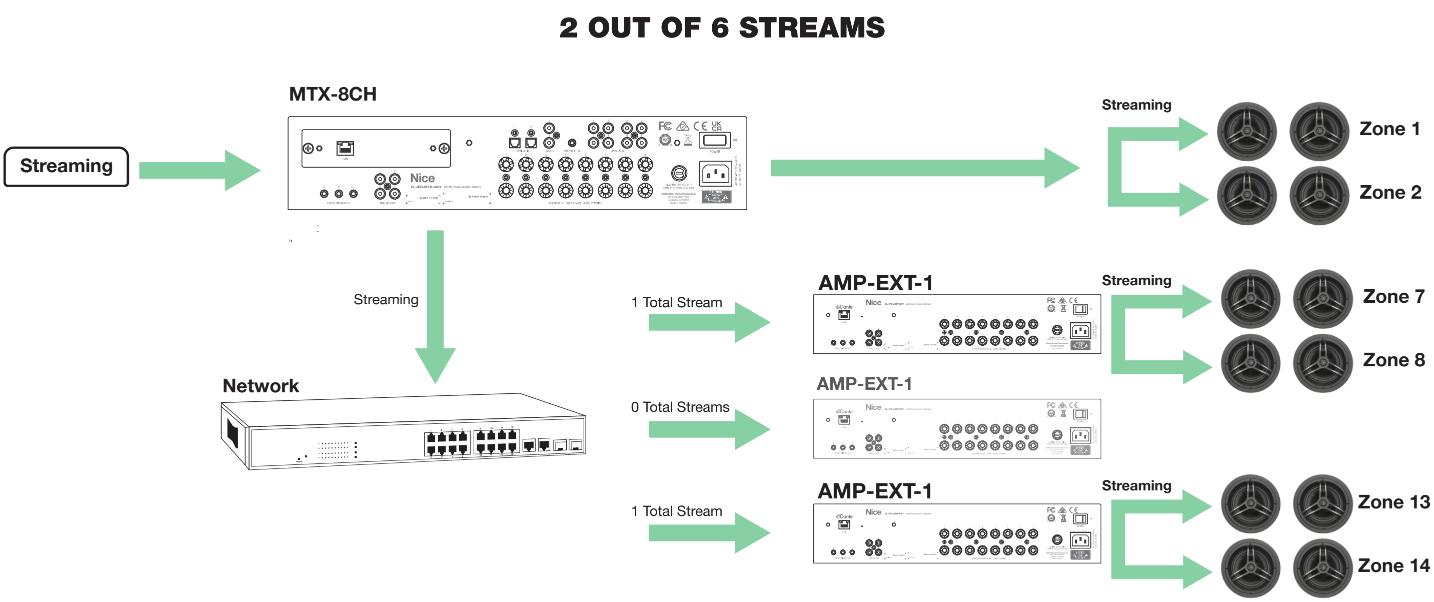
When a chassis requests a non-internal source, (that would be all sources for the EL-IPD-AMP-EXT) it requests the source from the hosting chassis. If the hosting chassis has an available transmit stream, it will then send the source to the chassis, using one of the available streams. The chassis can then internally distribute the source as it sees fit. For instance, if the chassis has 4 internal zones, and 2 of which request the same source. The source is sent to that chassis once, thus using 1 of its available streams, not 2. If another chassis requests the same source, that source is sent again to the new chassis, using another stream for the same source. So each source can use more than 1 stream depending on its use. As a system gets larger, it is much more efficient to send audio via Multicast – see Advanced section.

Chassis unicast sending capabilities:

- EL-IPD-MTX-8CH – 6 stereo streams
- EL-IPD-PRE-MSI – 8 stereo streams
- EL-IPD-PRE-SSI – 2 stereo streams

Example 1: Sending the same source to 2 different chassis

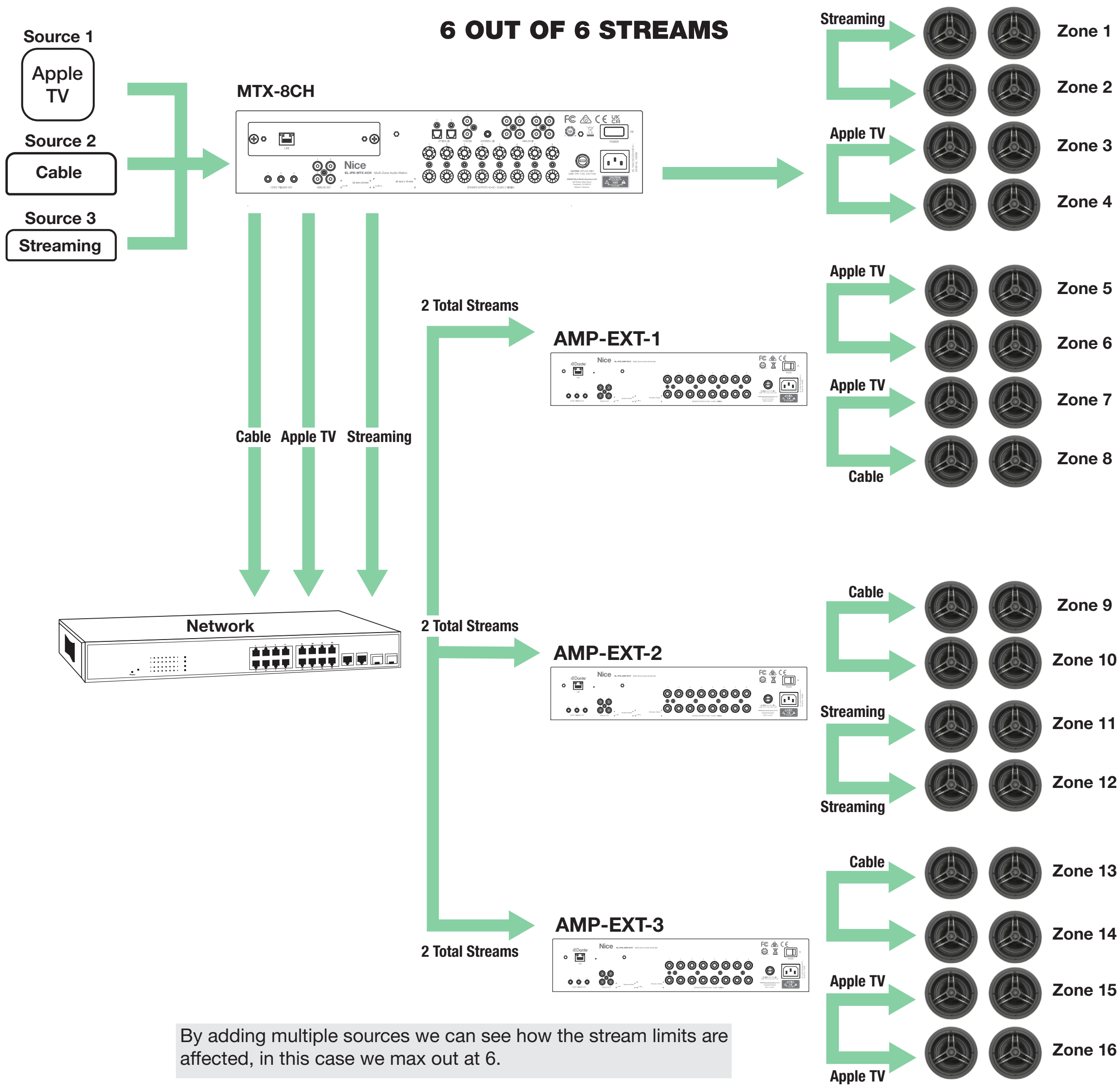
In this example, we show that sending the same source to 2 different chassis counts as separate streams.



In this example, the **Source 1 (Streaming)** is sent to each *chassis* that requested audio.

Example 2: Sending multiple sources

In this example, you can see we have hit the limit of available streams.



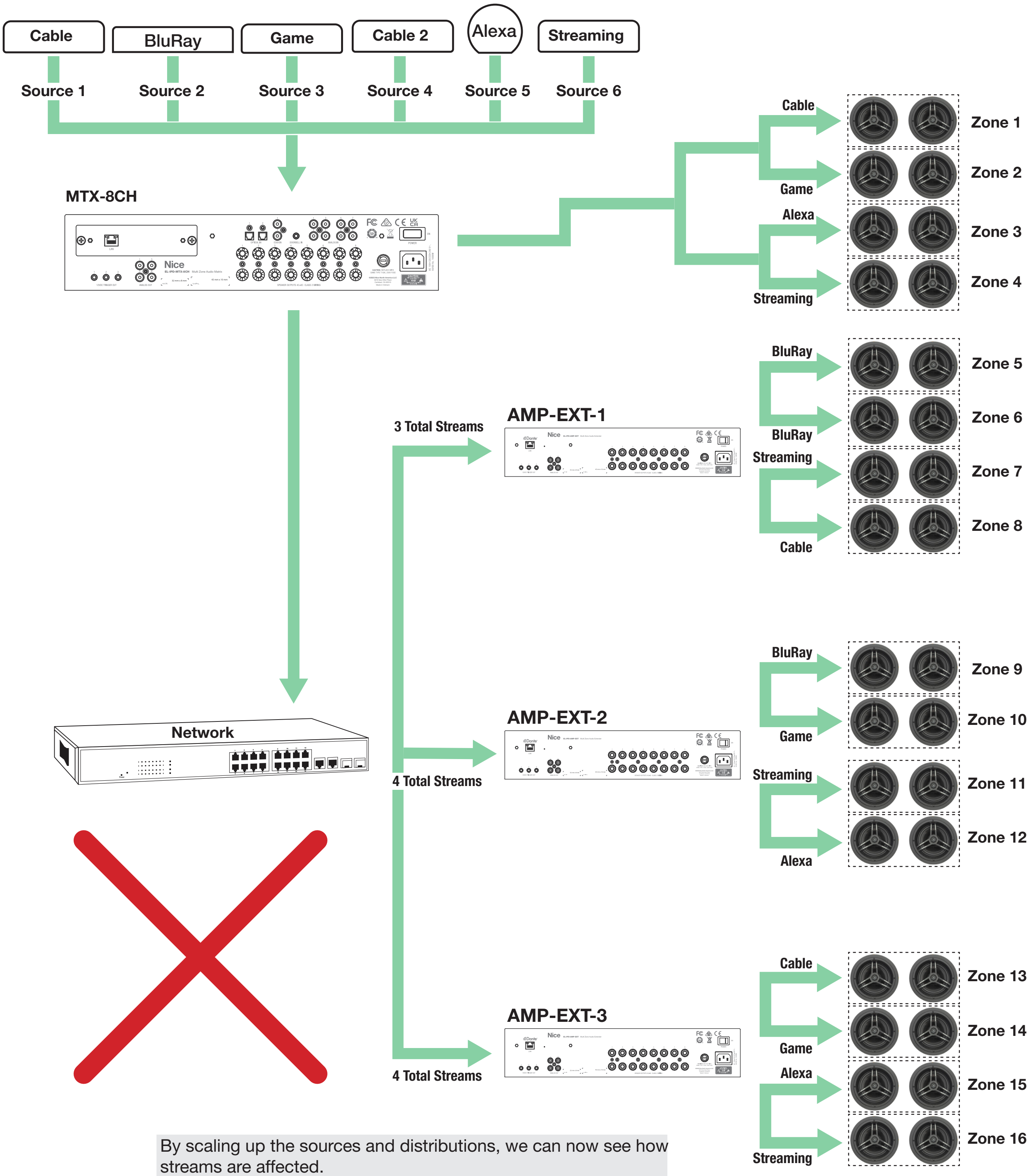
By adding multiple sources we can see how the stream limits are affected, in this case we max out at 6.

Example 3: Streaming past the limits

In this example, we are showing an extreme case of how many streams saturating each chassis would require

11 OUT OF 6 STREAMS

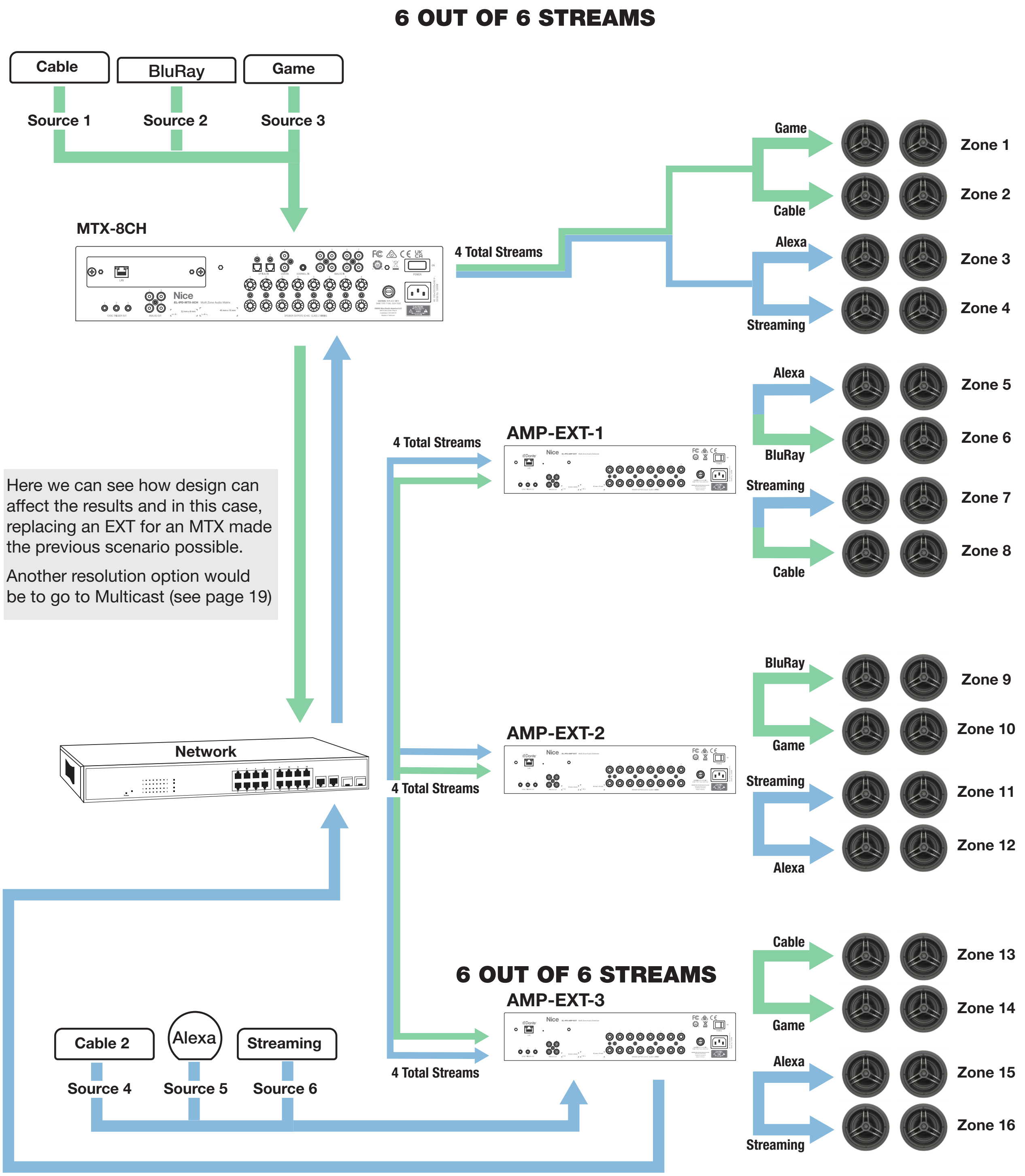
WARNING: will NOT work. Refer to the resolution displayed on the next page.



By scaling up the sources and distributions, we can now see how streams are affected.

Example 4: Designing around the limits

Adding a EL-IPD-MTX-8CH or EL-IPD-PRE-MSI to increase the streaming ability



Unicast network settings:

- Disable Green Ethernet, if applicable
- Do not use a EEE switch
- 10/1000 (gigabit) network switch

Common Mistakes:

- Having Multicast devices, most notably video over IP, on the same switch/subnet as an Nice Home Management IP audio chassis running in unicast.
- Cascading switches degrading network performance - see “Best Practices” (found on page 18).
- Hitting/exceeding the stream limits. When in Unicast (default) mode, be certain to plan the design before deployment.

Advanced Multiple Chassis Configuration – Dante Multicast

Multicast Overview

Similar to how VIP operates, each source can be sent to multiple endpoints using only one of the available streams. This is much more efficient. It is unlike Unicast, which needs to send the same stream to each chassis separately using another stream and increasing network traffic. However, Multicast does require specific networking gear that can handle the necessary configuration (limit of 64 zones).

Multicast is needed on larger systems because it can handle 5+ chassis, as opposed to the four in which Unicast is suited (design dependent). If you have an eight chassis **Multi-Chassis** system and would like to simultaneously send the same three sources to each chassis, three streams will be used. With Unicast, this would require 21 streams, therefore it could not be accomplished. This is also known as *Fanout*.

Hardware requirements:

- Layer 2/3 switches for all connected IPD devices

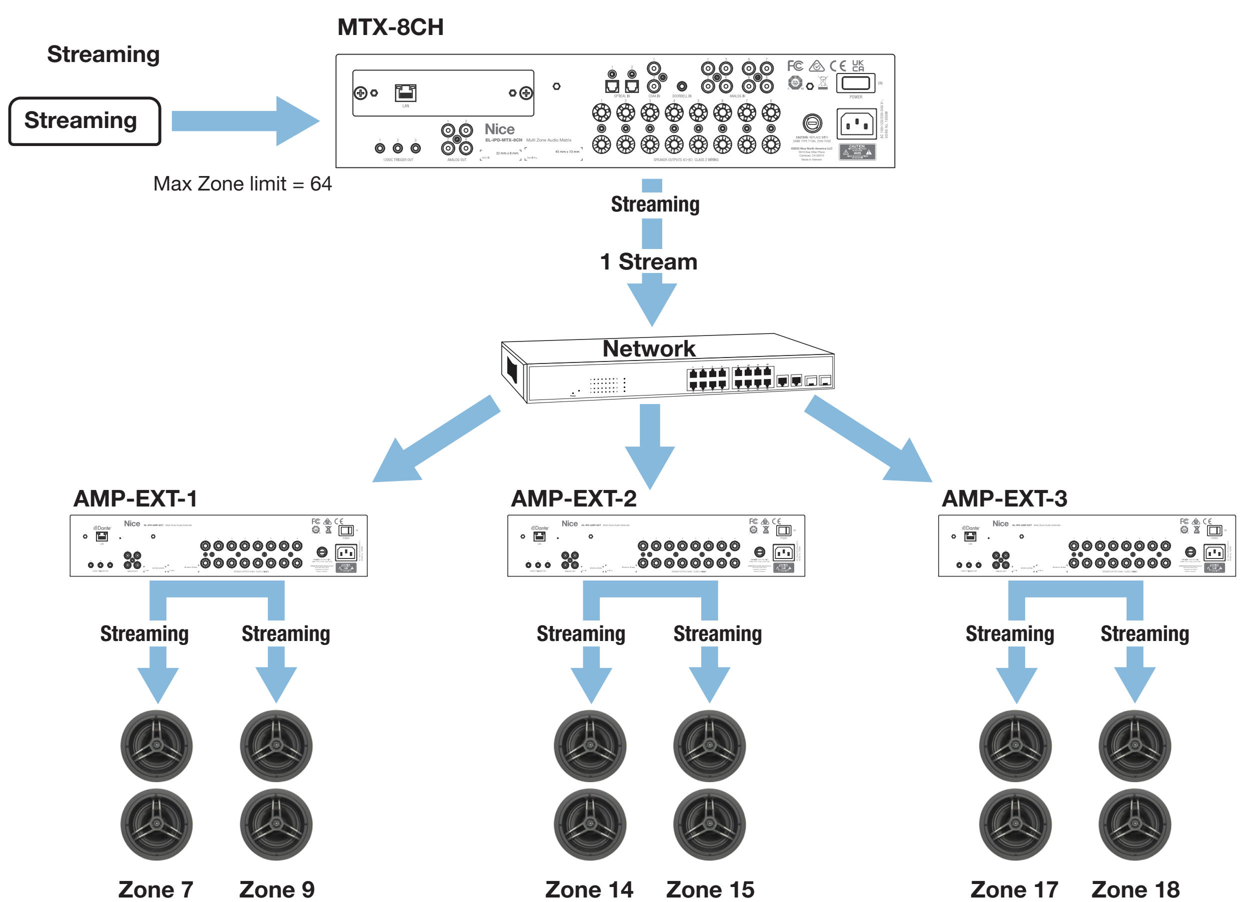
Chassis Multicast Sending Capabilities:

- EL-IPD-MTX-8CH – 12 stereo streams
- EL-IPD-PRE-MSI – 16 stereo streams
- EL-IPD-PRE-SSI/SIO – 4 stereo streams

Example 1: One source sent across 3 extender chassis'

In this example, one source can be sent to all necessary chassis and only uses one available stream. The Unicast counterpart, although much easier/quicker to deploy, would use three of the available streams.

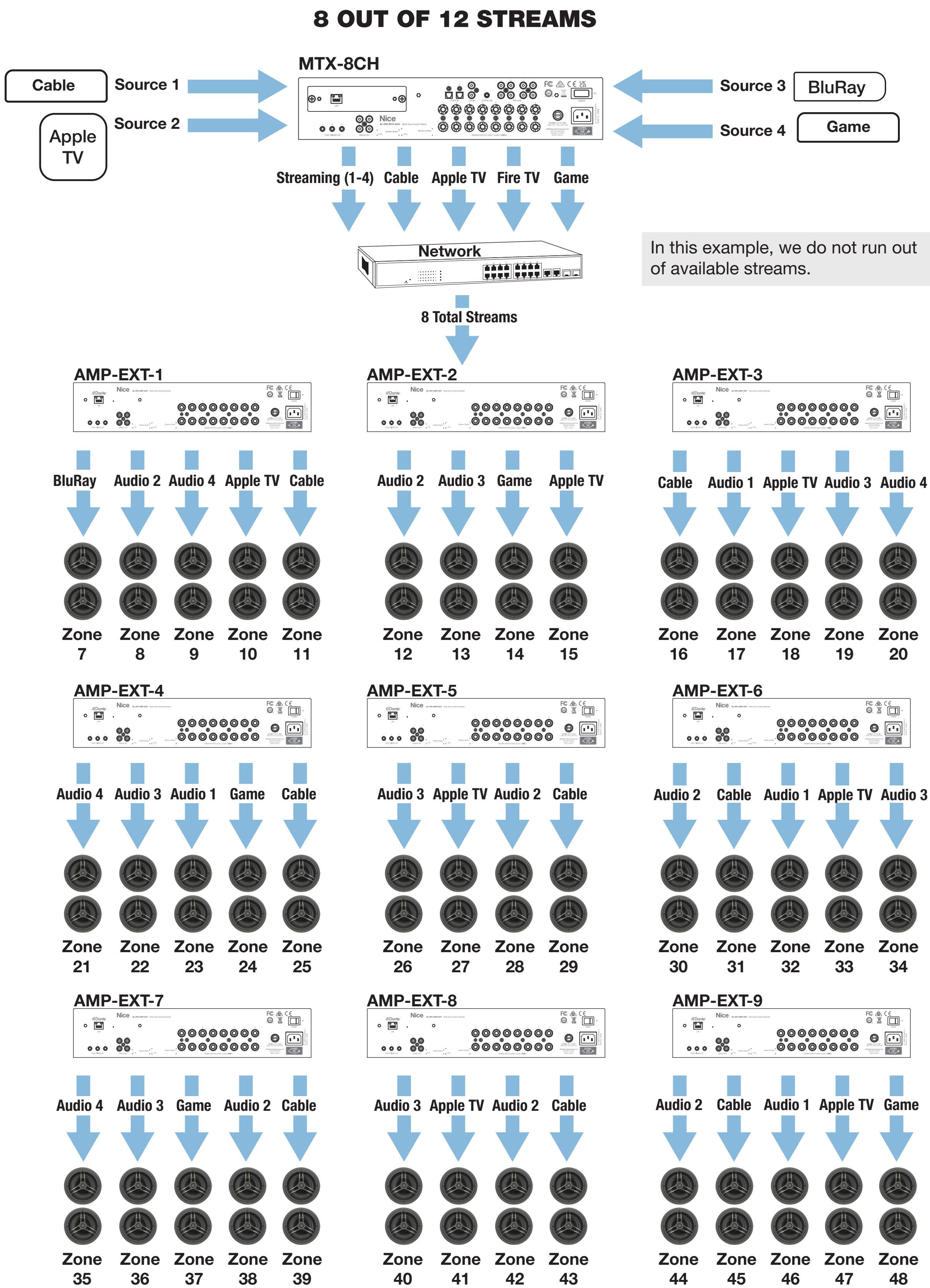
1 OUT OF 12 STREAMS



In this example, we send Streaming to three AMP-EXT chassis. In Unicast, this would require three streams to accomplish. But in Multicast, each source only takes one stream, no matter how many chassis receive.

Example 2: Eight sources across 9 extender chassis’

In this example, we can see that this is easily possible using Multi-cast. If in Unicast, this would require more than 40 streams and would inhibit network bandwidth.



NOTE: You can see Multicast is a better way to distribute the source streams but does require network gear that are able to handle multicast traffic as well as network configuration.

Multicast Settings:

Required on **all** switches/ports across the network with an **Nice Home Management IP Amp** plugged in.

- IGMP Snooping On
- Unregistered Flooding On & Fast Leave On
- Querier set to the root switches IP(same IP used for all switches querier)

If using a single switch network or stacked switches, set the querier to its address. If using multiple switches, chose a master switch and set them all to this address. In this configuration an access point must be placed on a separate switch.

- Fast Leave On
- QoS DSCP Setup

Common Issues

Wi-Fi issues

When a Wi-Fi access point is connected to the same switch as an Nice Home Management IP Amp in Multicast mode, network routing can flood the Wi-Fi and degrade the performance of the link if the hardware and/or network settings are not adhered to. You may be experiencing this condition if you are experiencing loss of the Wi-Fi signal. There are two ways to resolve this issue:

Option #1: Set the network to handle the multicast traffic appropriately.

1. Temporarily power down any Nice Home Management IP Amps and EL-IPD-PRE-SSI/SIO.
2. With the network congestion removed, you can now make the network settings changes.
3. Follow the correct network switch settings for Multicast use on all switches.
4. Follow the correct network switch settings for multicast use on all switches in the project.
5. After the changes have been completed, you can power on the IPD devices.

Option #2: Factory default the IPD chassis (only applicable with no EL-IPD-PRE-SSI/SIO's set to Multicast)

This method is used when non appropriate network gear is used and thus not able to run multicast.

- Press and hold the micro reset button (on the Dante card, or next to the Ethernet connection) until a click is heard (~ 30 seconds). This will reset the chassis back to factory condition.

IMPORTANT: While the chassis is rebooting and before writing down the chassis configuration, uncheck the Multicast box.

- The chassis will need to be written to from the configurator after it comes back online to sync it with the zone settings.
- Make any changes necessary to the network before re-enabling Multicast mode.

Common Multicast issues:

- **No/intermittent Wi-Fi**: incorrect switch setting or non-compatible hardware
- **Improper multicast network setup**: IGMP & QoS. See above for Luxul specific settings.
- **Incorrect network hardware**: To use multicast, the network must be able to handle IGMP and QoS which is found on layer 2/3 switches.
- **Poor network infrastructure**: Poor cascading, ends, cables

EXAMPLE: An EL-IP-PRE-SSI, for example, is often connected to a small switch located behind the TV which is not layer 2/3. If the connected EL-IP-PRE-SSI is in Multicast mode as the switch cannot handle the traffic and as such will cause flood the switch as well as any other devices connected to that switch. Use a layer 2/3 switch or run another wire for the EL-IPD-PRE-SSI/SIO.

Best Practices

Network setup and deployment: (unicast and multicast)

Networking can play a large part in installations, esp as they expand. For proper deployment and system expectations, the below are a few of the best practices when designing/deploying a Dante setup. As Dante is sending timed audio through the network, the network must be able to handle that bandwidth and routing, as well as the packet timing appropriately.

Use Gigabit switches

- Each stream can use ~6-10 Mb. You can only use ~60% of the available space on the port:
- A 10/100 port can support 6 streams before overwhelming the port
- A 10/1000 port can support the full amount of supported streams

Do not connect an Nice Home Management IP Amp device directly to the router

- Often, especially ISP provided, a router does not have the ability to route the traffic for Dante appropriately

Connect all Nice Home Management IP Amp devices to the same switch, if at all possible

- Best approach, most reliable
- This will keep the audio traffic internal to only that switch
- Will prevent the degradation that occurs from cascading via copper

Cascading Switches

- A Gigabit Cat-6 copper link can handle 800-950MB average, If that switch is also handling WiFi or streaming, it can run out of bandwidth pretty fast
- Best approach – fiber link switches if possible
- Alternate - Can multi-link a switch via copper. Do not cascade with a single CAT-5E
- Do not use a EEE switch and be sure green Ethernet is disabled on class 2/3 switches

EEE and green Ethernet is not supported by Dante and will not operate correctly. On some layer 1 switches, it is not possible to turn this off so make sure that the switch is compatible first!

Firmware Updating

- Use the pre-installed NIC card on the EL-IPD-MTX-8CH & EL-IPD-PRE-MSI unit! The firmware can take 15 minutes
- Be sure all zones are off before commissioning an update
- Do not disturb the chassis after you start the update

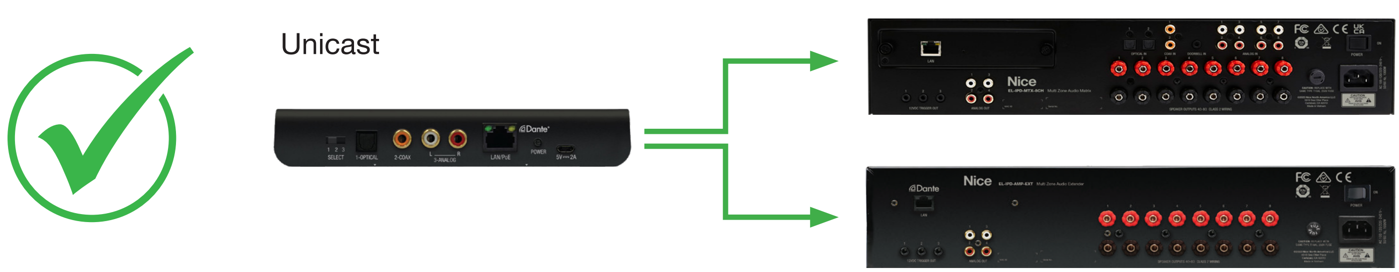
Plan the zone outputs to best suit you needs

Now that you have an understanding of the audio streams, you can in some cases, reduce the streams by designing the zone layout differently.

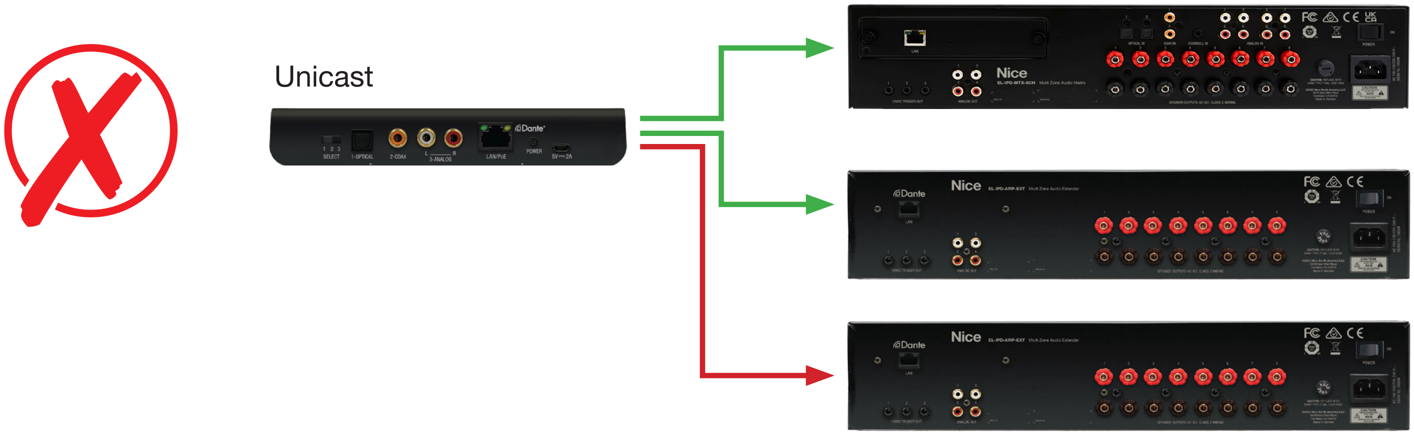
In most cases, you won't be sending every audio source to every chassis simultaneously. You can potentially design around creating fewer streams to gain a larger footprint in unicast mode.

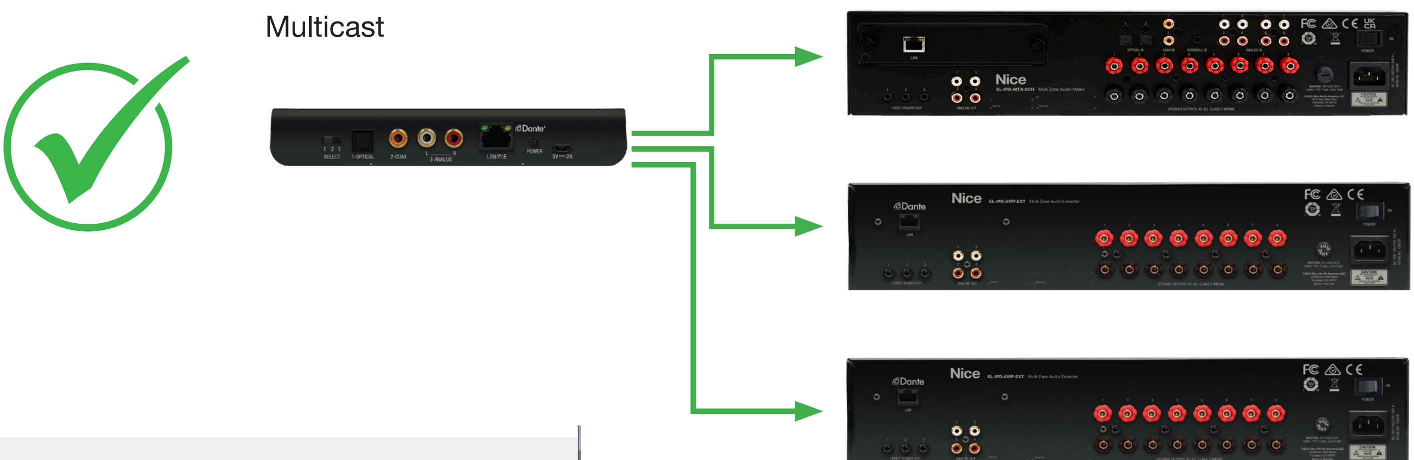
Plan EL-IPD-PRE-SSI use appropriately

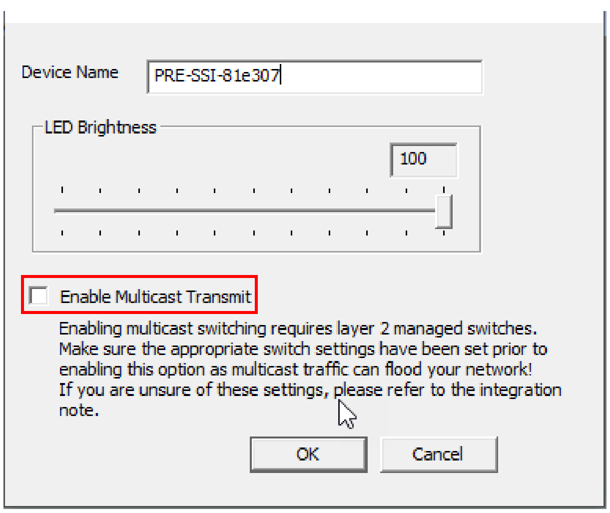
In unicast, the EL-IPD-PRE-SSI is limited to sending two output streams simultaneously. In other words, it will send the source to 2 chassis at the same time. If you want to stay in unicast, plan the use appropriately.



NOTE: Can run in Unicast even if the other chassis' are running Multicast.







NOTE: Once set to Multicast, proper network gear and settings are required in order to return this setting back Unicast.

Low power mode

- If turned on, this will cause a zone to take up to 20 seconds to turn on, but will save power
- This will render the doorbell, if equipped, not trigger
- Know the front panel L.E.D. color code as this will show if the chassis is in low power mode

When do I need multicast?

- When you will exceed the stream limitations, typically over 4 chassis (but depends on system use/design)
- When used with other multicast traffic, such as MOIP
- When sending an EL-IPD-PRE-SSI to more than 2 chassis simultaneously

Doorbell

When the doorbell is triggered, a lot has to happen in the background and thus can only effectively trigger 6 zones on a doorbell chime in the time allotted or the page may be ending by the time the zone is setup.

You can reduce the operation to make the chimes happen as fast as possible by

- Set the delay to 0.0 in the intercom paging settings
- Not selecting “Volume control” option in the intercom page settings

Latency

- We have a 2ms window for allowed latency. Default Latency is 2ms
  If you require more, there is a drop-down option you can use
- Audio will mute if the latency is greater than the set latency
- Caused by network bandwidth, cascading, poor infrastructure
- Switch configuration, ends and even patch cables make a difference

Common Mistakes:

- Inopportune system design which can play a large part on the functionality
- Overloading a cascaded switch link
- Overloading a 10/100 switch port
- Green Ethernet enabled or EEE switch
- Connecting chassis directly to a ISP provided router
- Not updating firmware

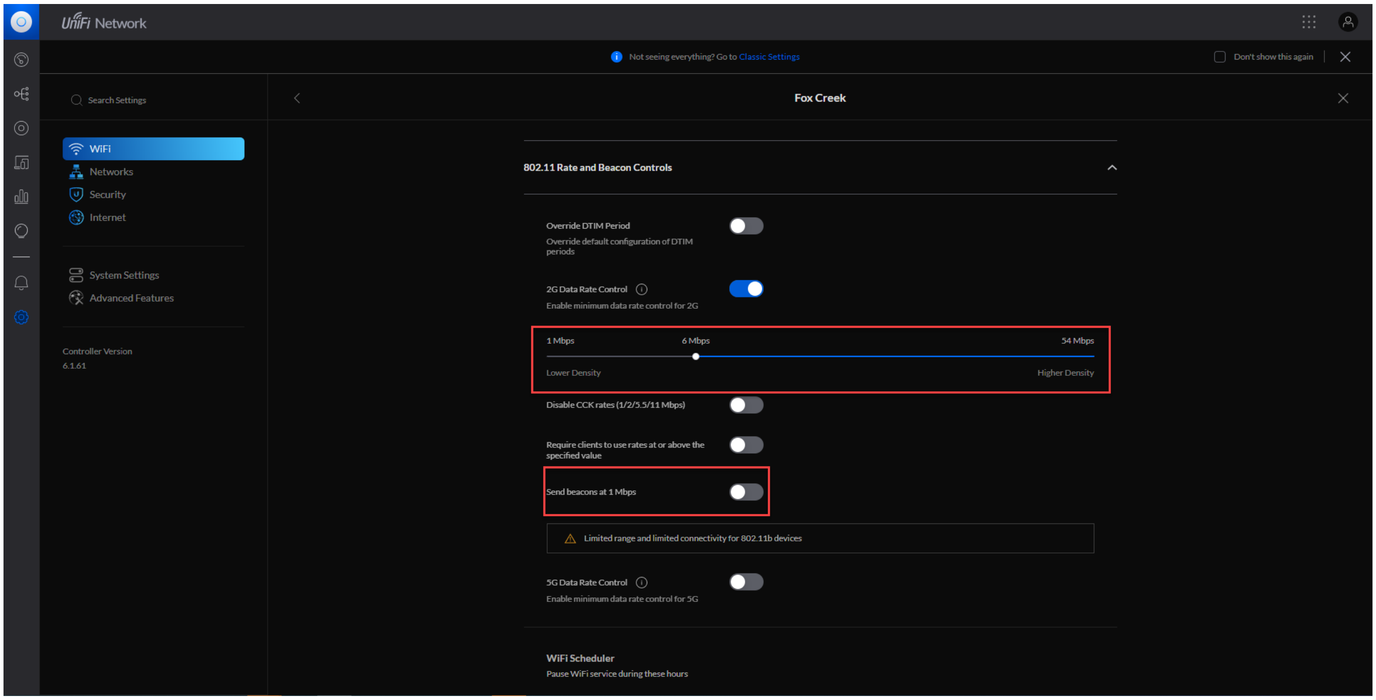
Networking Details

In this section, we will list information and specifics on network switch brands, as well as a sample of Luxul's Multicast settings.

Ubiquity

For Multicast deployments, only the USW-EnterpriseXG and USW-Pro series support multicast.

In Multicast, the slider needs to be put down to 1Mbs and beacon sending enabled for 802.11 devices to work properly.



Araknis

For Multicast Deployments, the DSCP should b set up in this manner:

- Set **DSCP 8** to **Queue 3**. This is for low priority "reserved" usage
- Set **DSCP 46** to **Queue 7**. This is for medium priority Audio traffic
- Set **DSCP 56** to **Queue 8**

Luxul

- For multicast deployments, use the required SW-610 or 510 series switches
- Luxul fw V1.0.3 (or higher) is required to set up a multicast deployment
- Be sure to set "Unregistered IPMCv4 Flooding" to ***Disabled*** on ports that are not connected to an IPD product

Configuring Dante

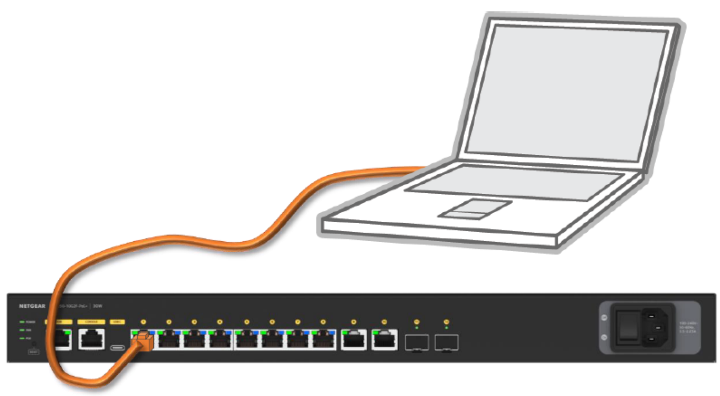
How to configure Dante audio on M4250 & M4300

M4250 & M4300 Configuration Guide for Dante Audio Devices

The functions of Dante devices rely on three key network technologies: Discovery and control (mDNS and known multicast group, QoS (IP DSCP to transmit critial packet, PTP), and Multicast Registration (IGMP for audio streams to join and leave a broadcast).

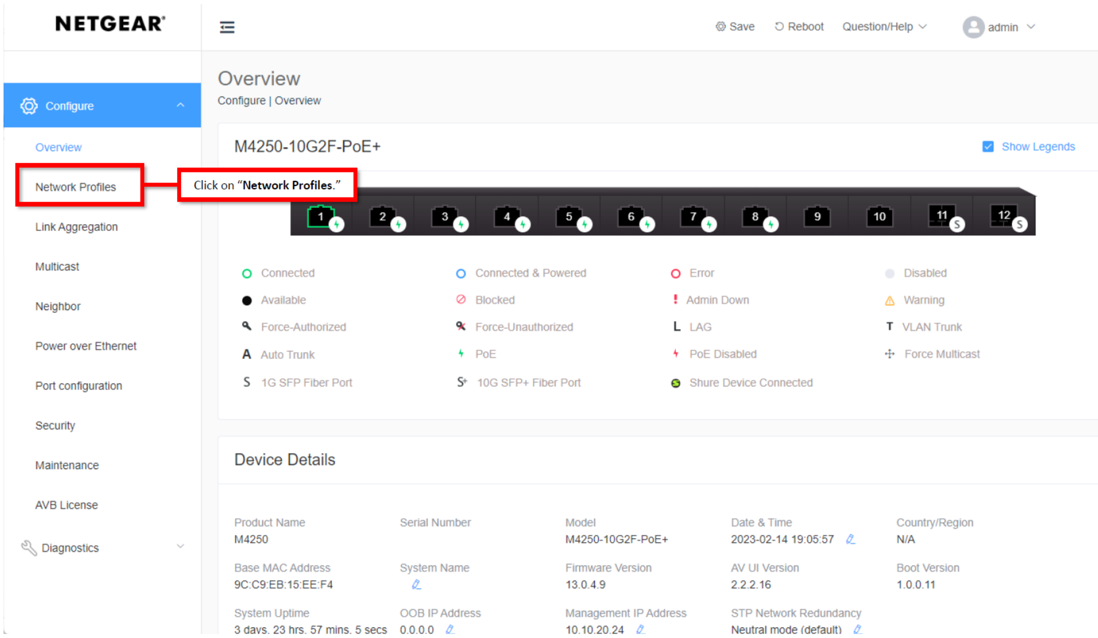
- Upgrade your switch to the latest firmware
- It is recommended to clear any previous configuration on the switch

Dante Audio Configuration Using the Default VLAN

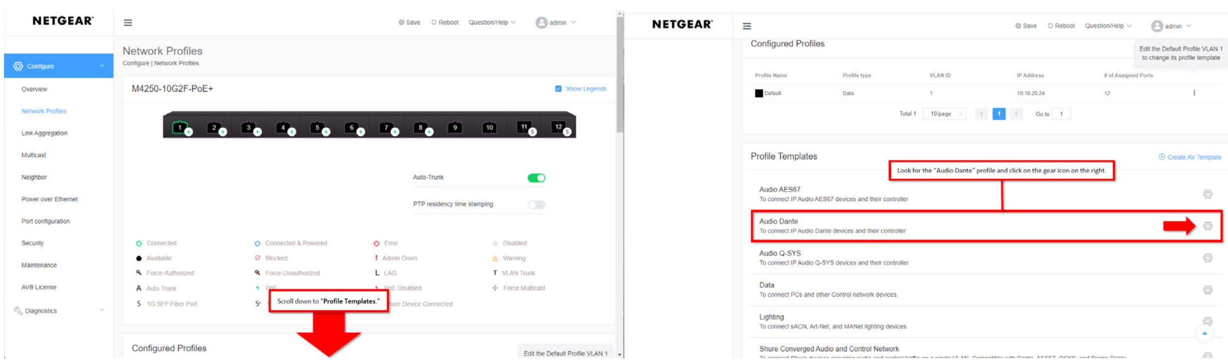


1. Log into your Netgear switch's AV UI:

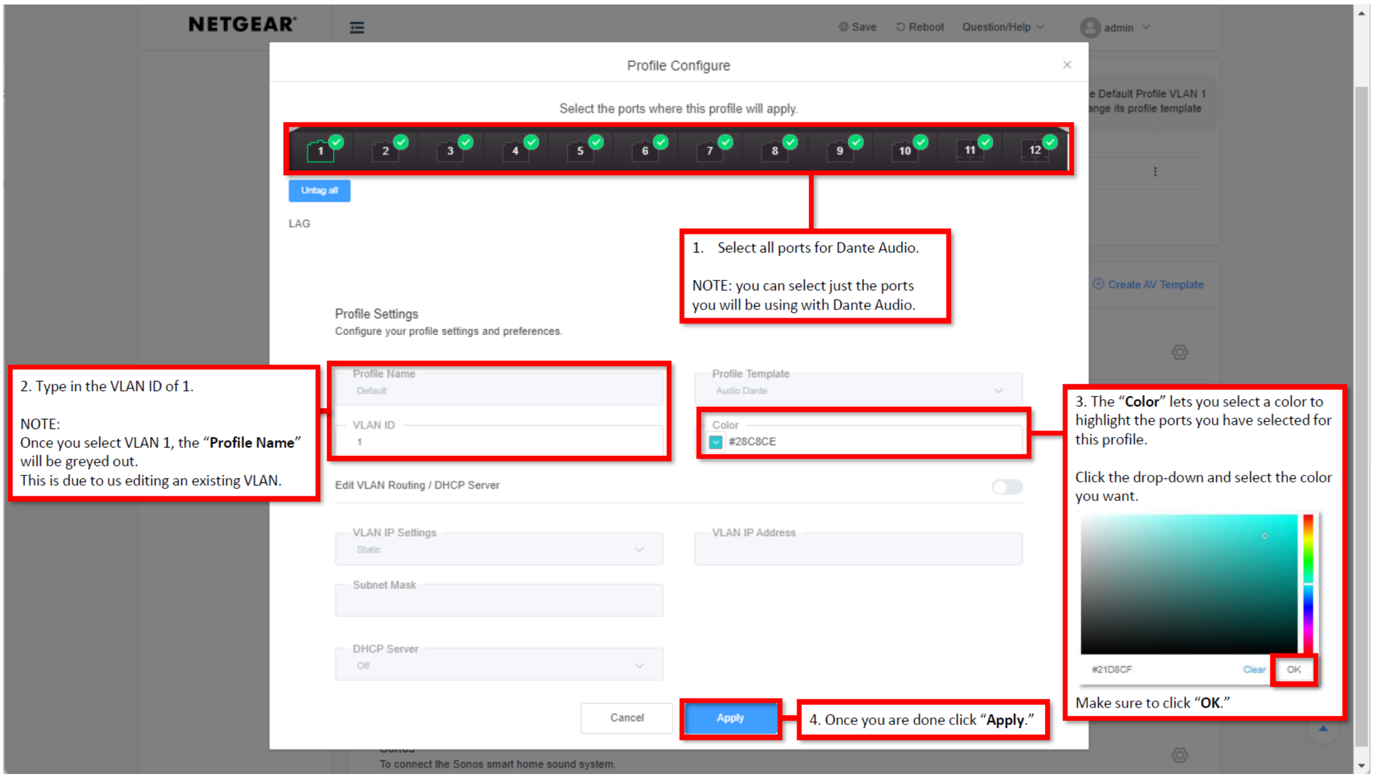
2. Click on Network Profiles



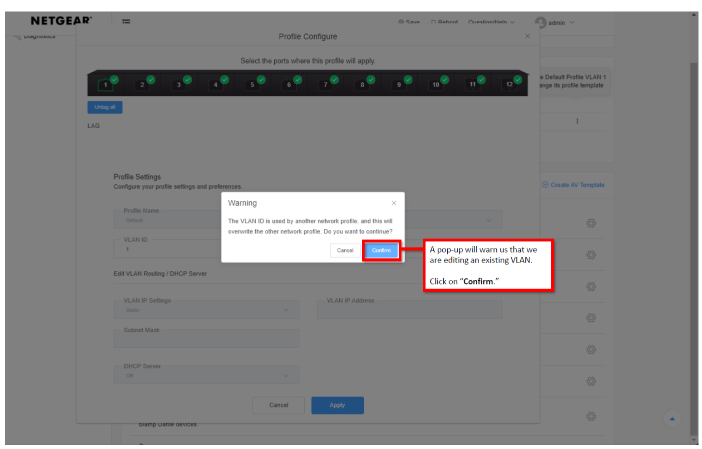
3. Scroll down to "Profile Templates" and press the gear next to "Audio Dante"



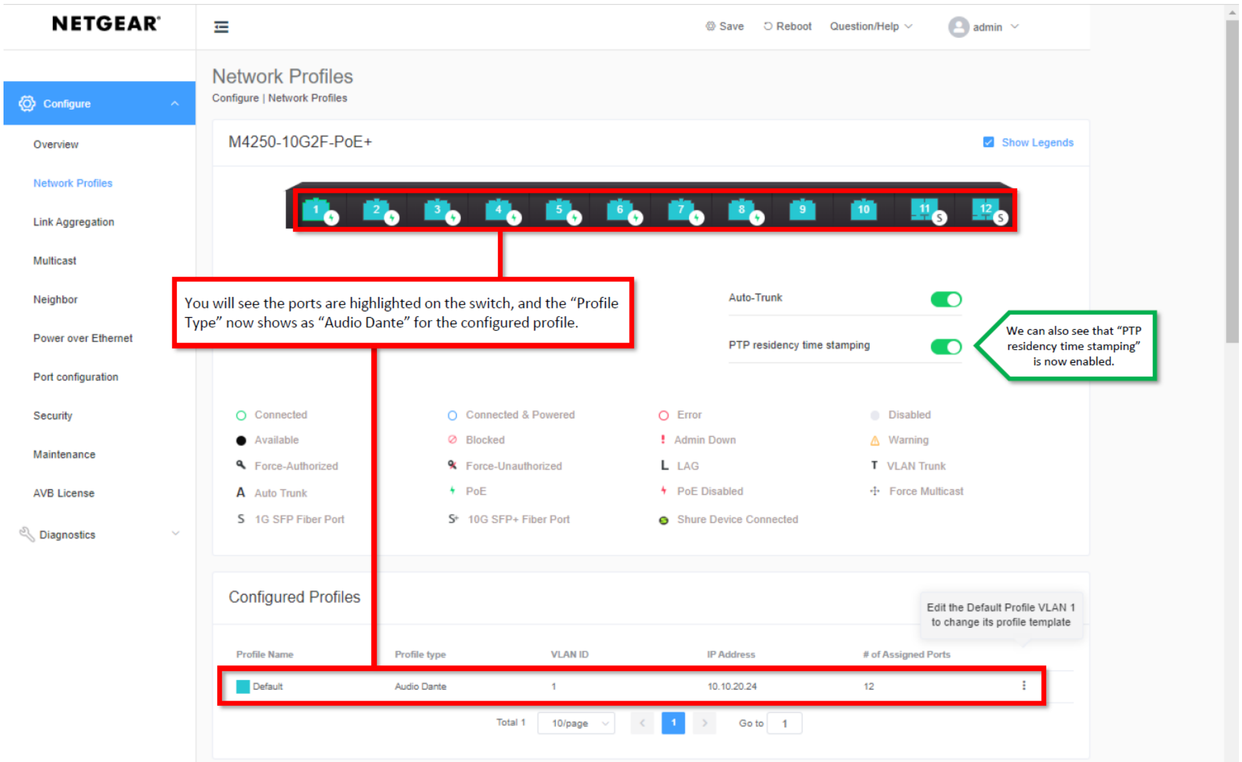
4. Make the following changes:
    a. Check all the ports with an IPD connected
    b. Make sure VLAN ID is 1
    c. You can select a color to indicate an IPD device is present under Color
    d. Click Apply



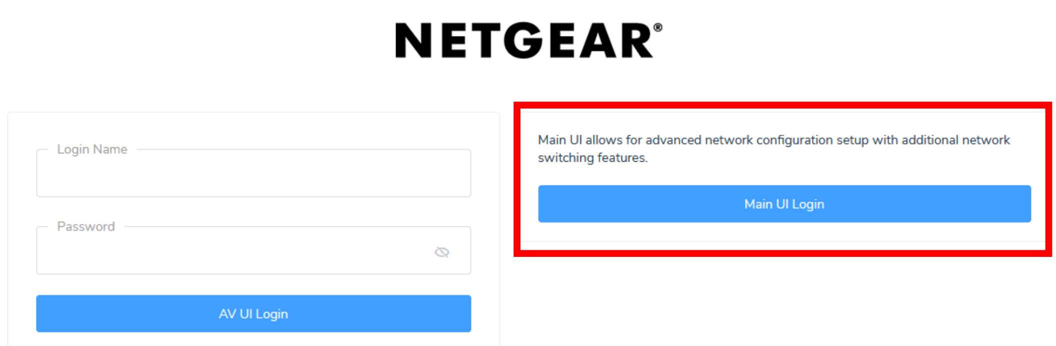
5. Click Confirm on the warning that comes up. The warning is because you are modifying an existing VLAN



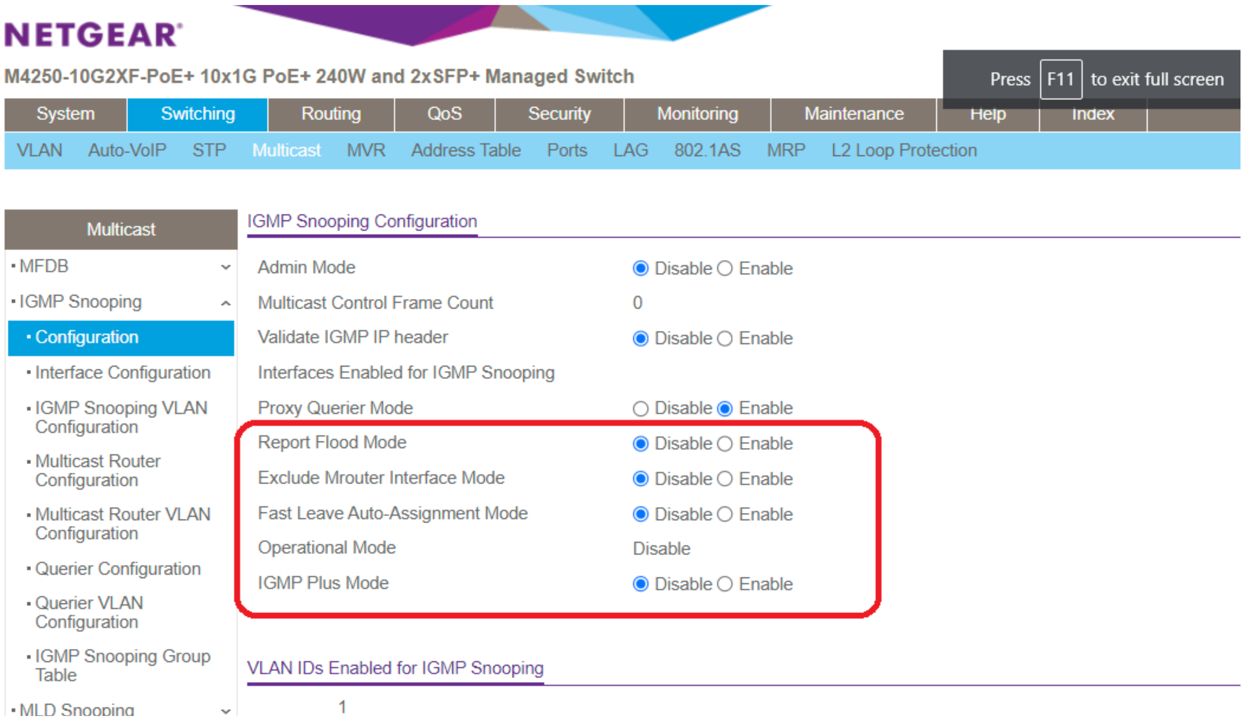
6. Confirm that for ports that have an IPD connected device:
    e. Indicate as Audio Dante
    f. PTP is enabled
    g. Show the optional color



7. Click save (upper right) then log out and
8. Now, Log in to the Main UI by selecting it from the screen below



9. Under Switching -> Multicast->IGMP Snooping->Configuration, Disable:
    h. Report Flood Mode
    i. Exclude Mrouter Interface Mode
    j. Fast Leave Auto-Assignment Mode
    k. GMP Plus Mode



10. Click save

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800-421-1587

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M – F, 6am – 4pm PST

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